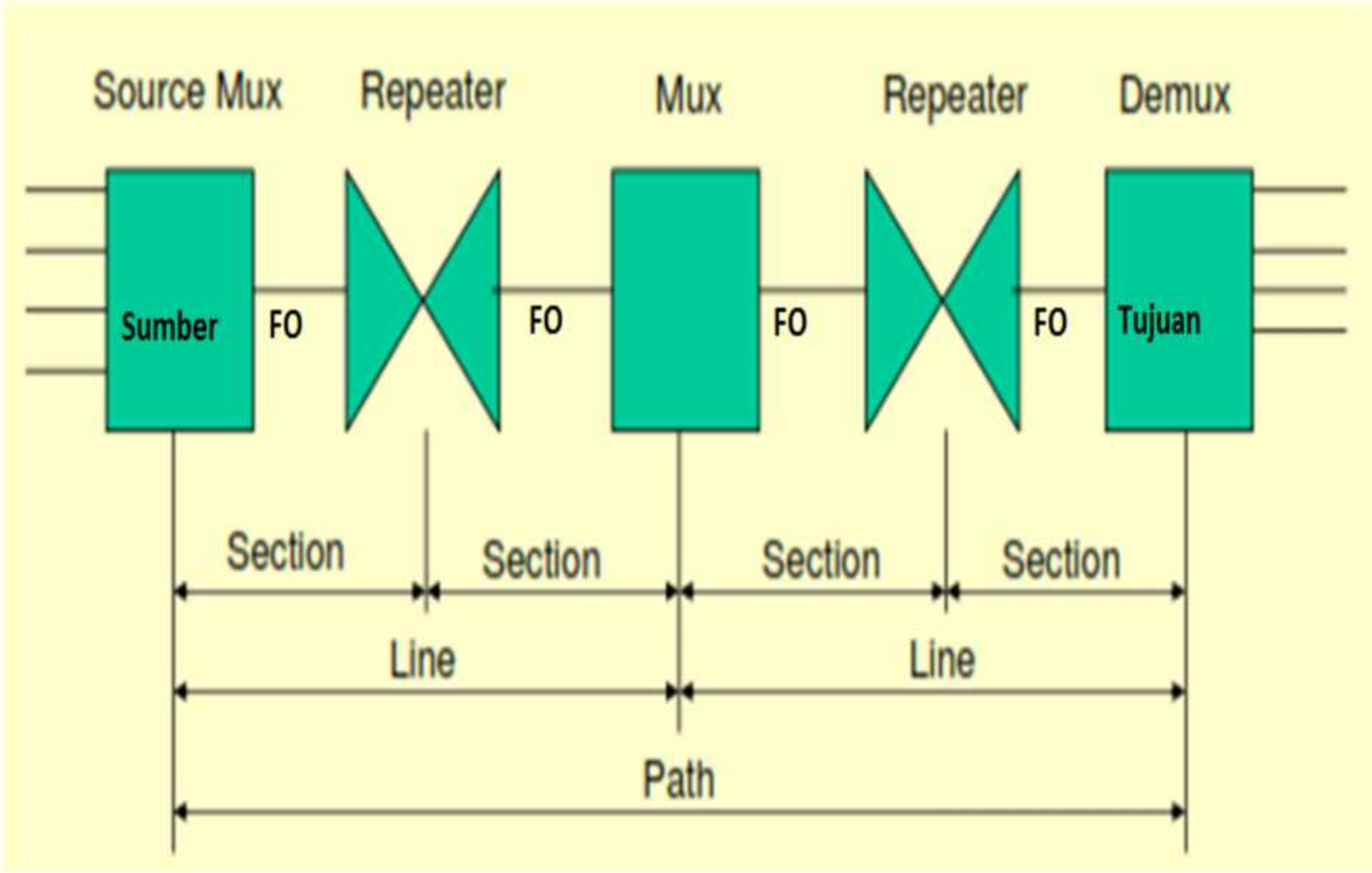
❖ SONET/SDH - Layering

- ➢ SONET/SDH system consists of switches, multiplexers and repeaters (and the fiber in between)
- ➢ PATH is the connection between source and destination
- ➢ LINE runs between two multiplexers (possibly through repeaters)
- ➢ SECTION is the connection of any two devices (point-to point)
- ➢ PHOTONIC is the e/o and o/e conversion layer

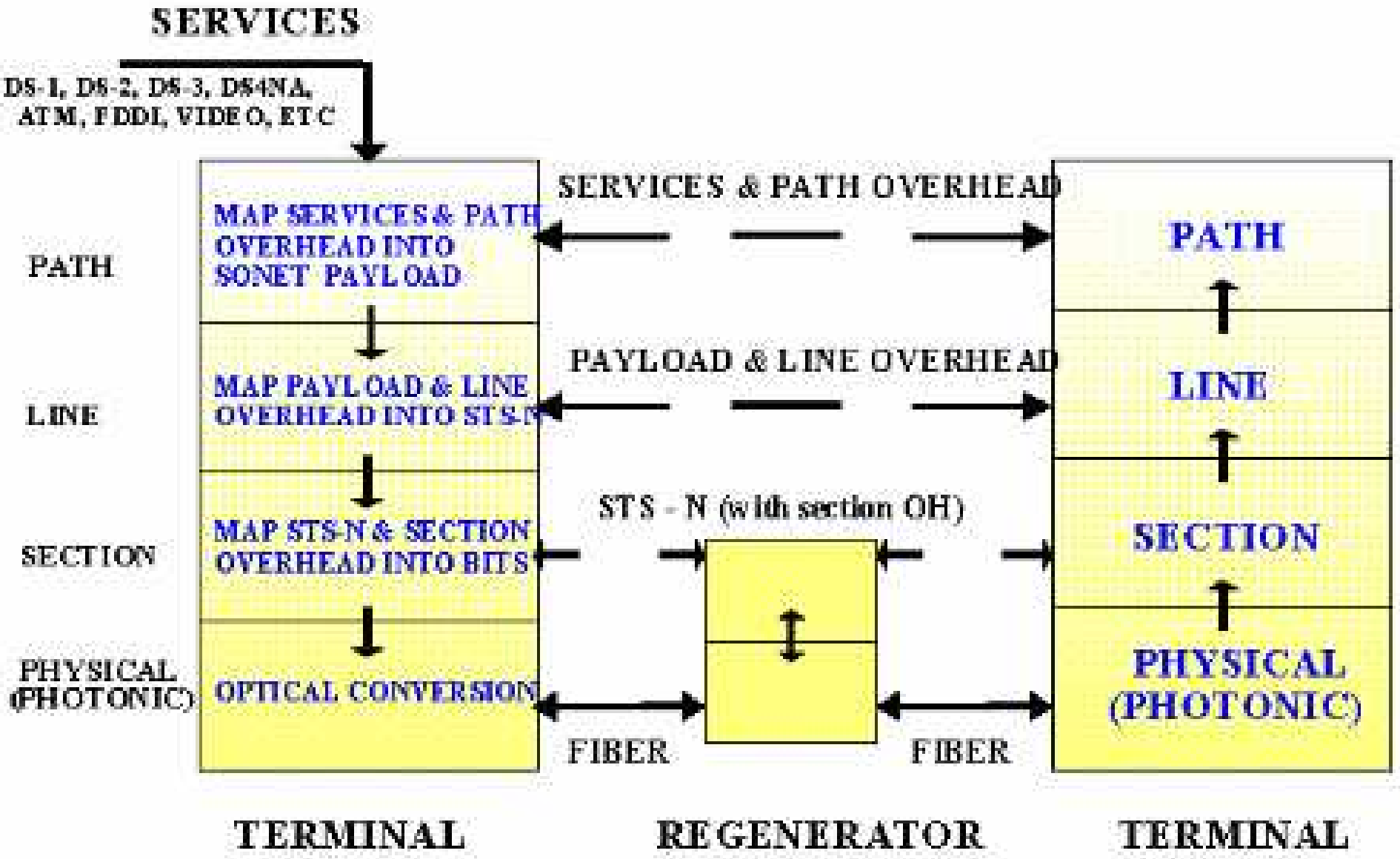
- e/o = electical / optical
- o/e = optical/ electrical

- SONET - Synchronous Optical NETwork
- SDH - Synchronous Digital Hierarchy (European standard equivalent to SONET)

SONET/SDH - Layering

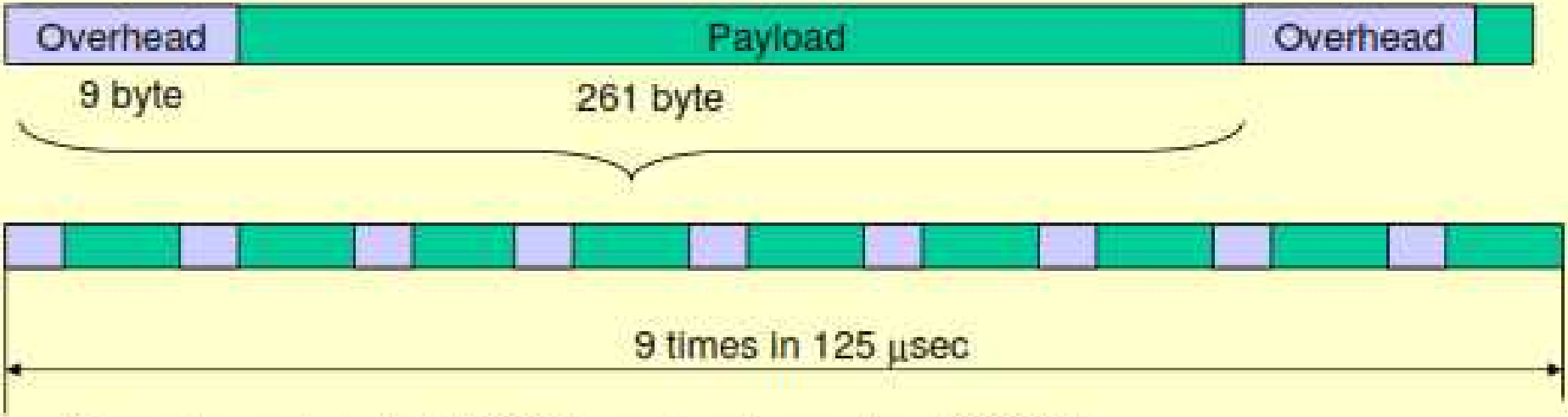


SONET/SDH - Layering



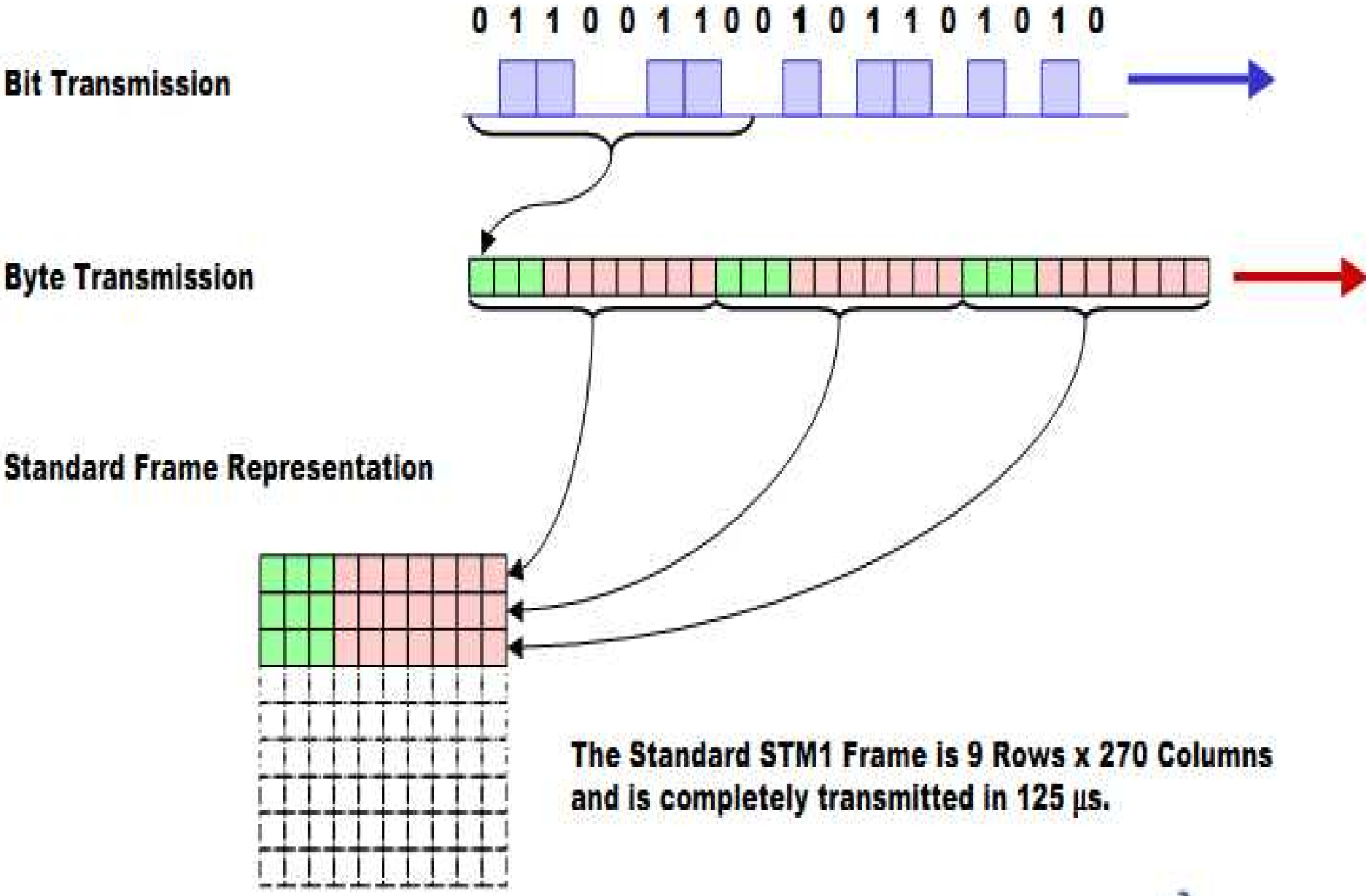
❖ SDH - Framing

- Frame format in SDH:

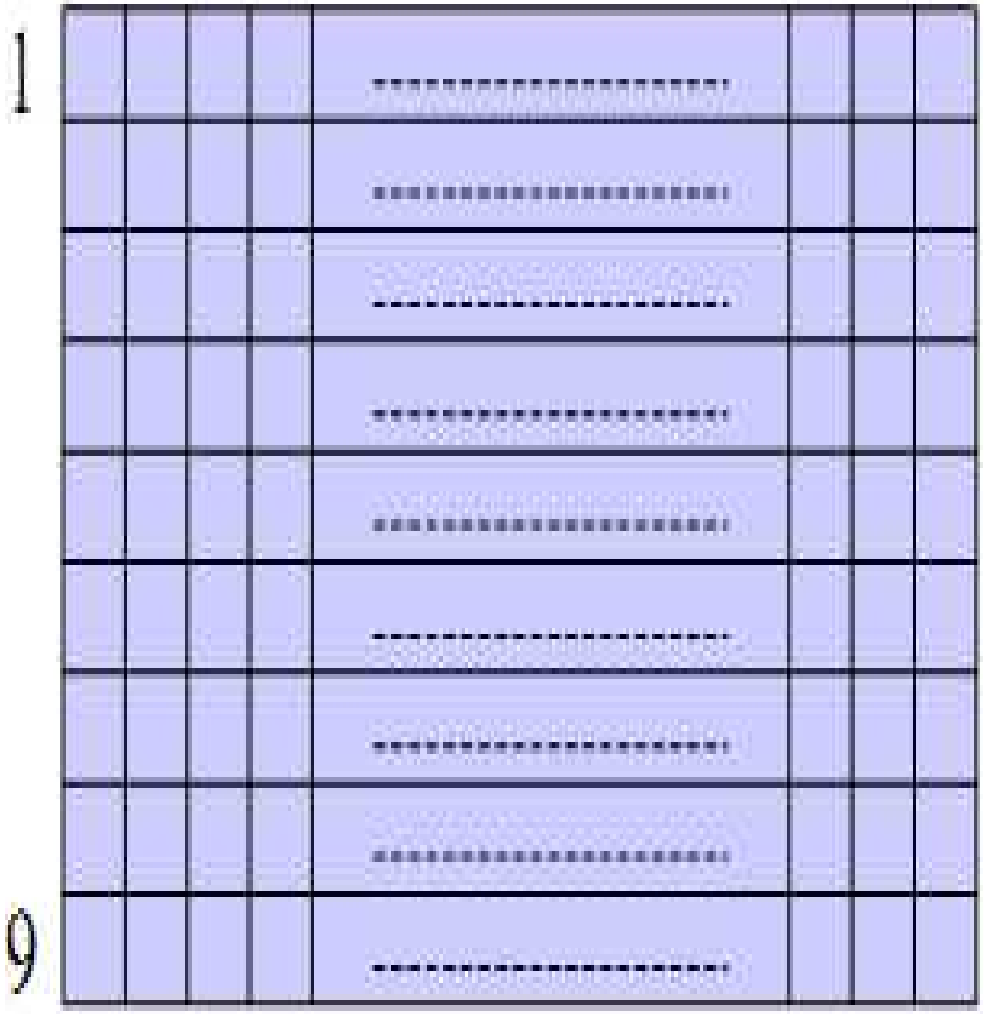


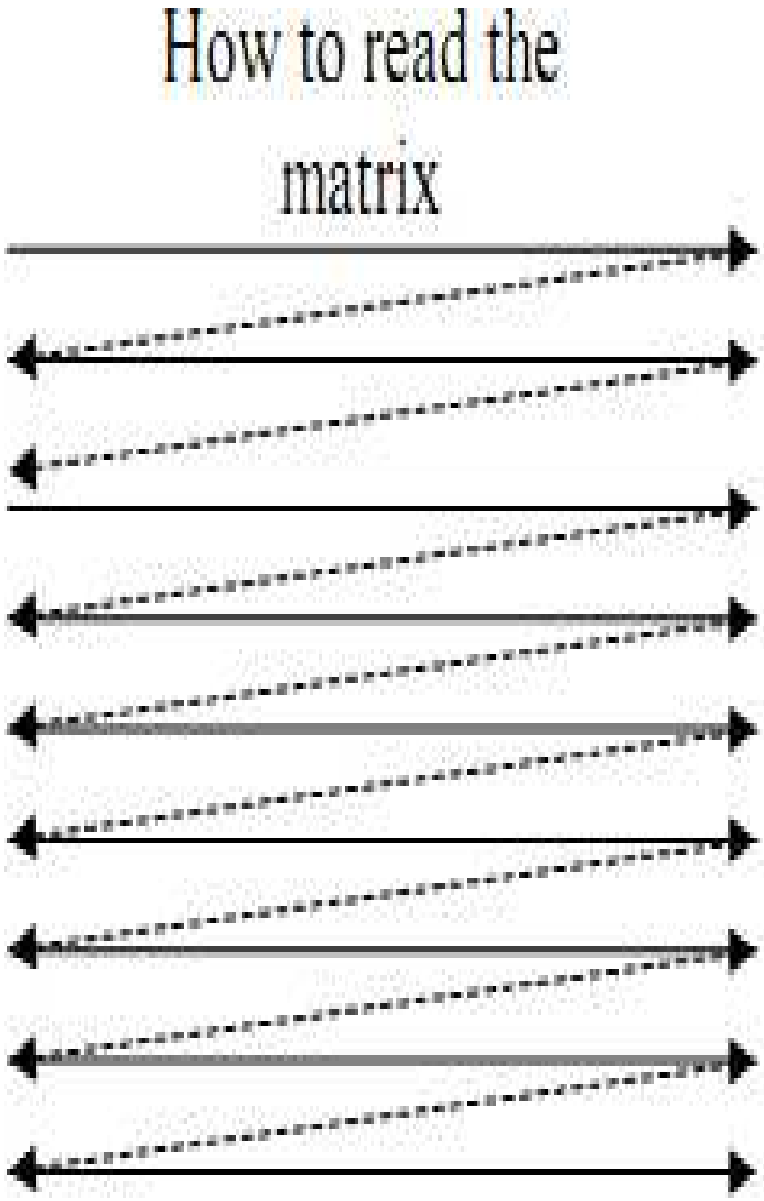
- Sampling rate of a 3400 Hz voice channel -> 8000 Hz
- 8000 Hz * 1 byte = 64 kbit/s
- 9 * 270 * 8 bit * 8000 Hz = 155.52 Mbit/s
- Basic data rate in SDH
- STM-1 -> Synchronous Transport Module

SDH - Framing

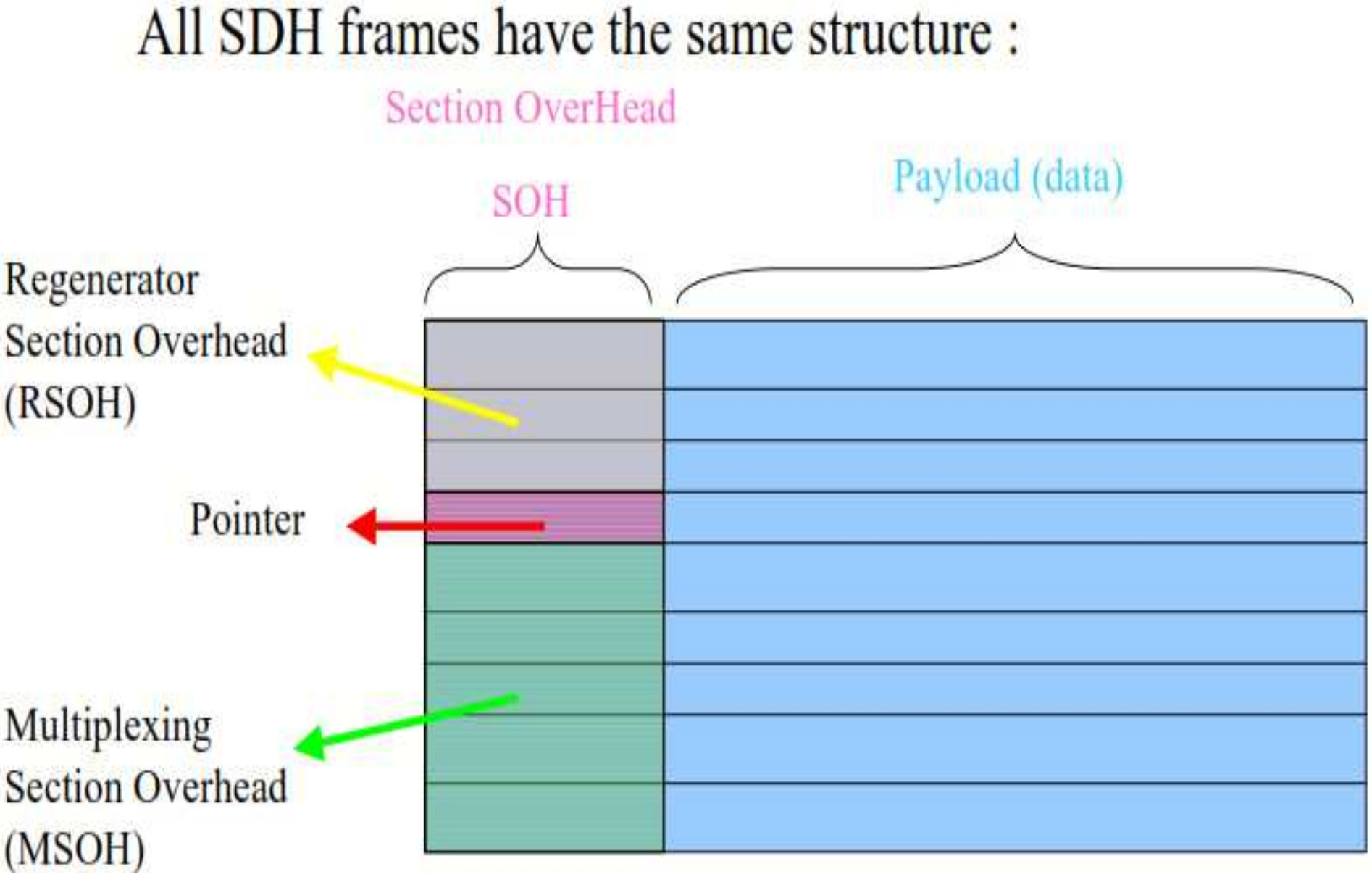

0 1 1 0 0 1 1 0 0 1 0 1 1 0 1 0 1 0
Bit Transmission
Byte Transmission
Standard Frame Representation
The Standard STM1 Frame is 9 Rows x 270 Columns
and is completely transmitted in 125 µs.

SDH - Framing





SDH - Framing


All SDH frames have the same structure :
Section OverHead
SOH
Payload (data)
Regenerator
Section Overhead
(RSOH)
Pointer
Multiplexing
Section Overhead
(MSOH)

SDH - Framing

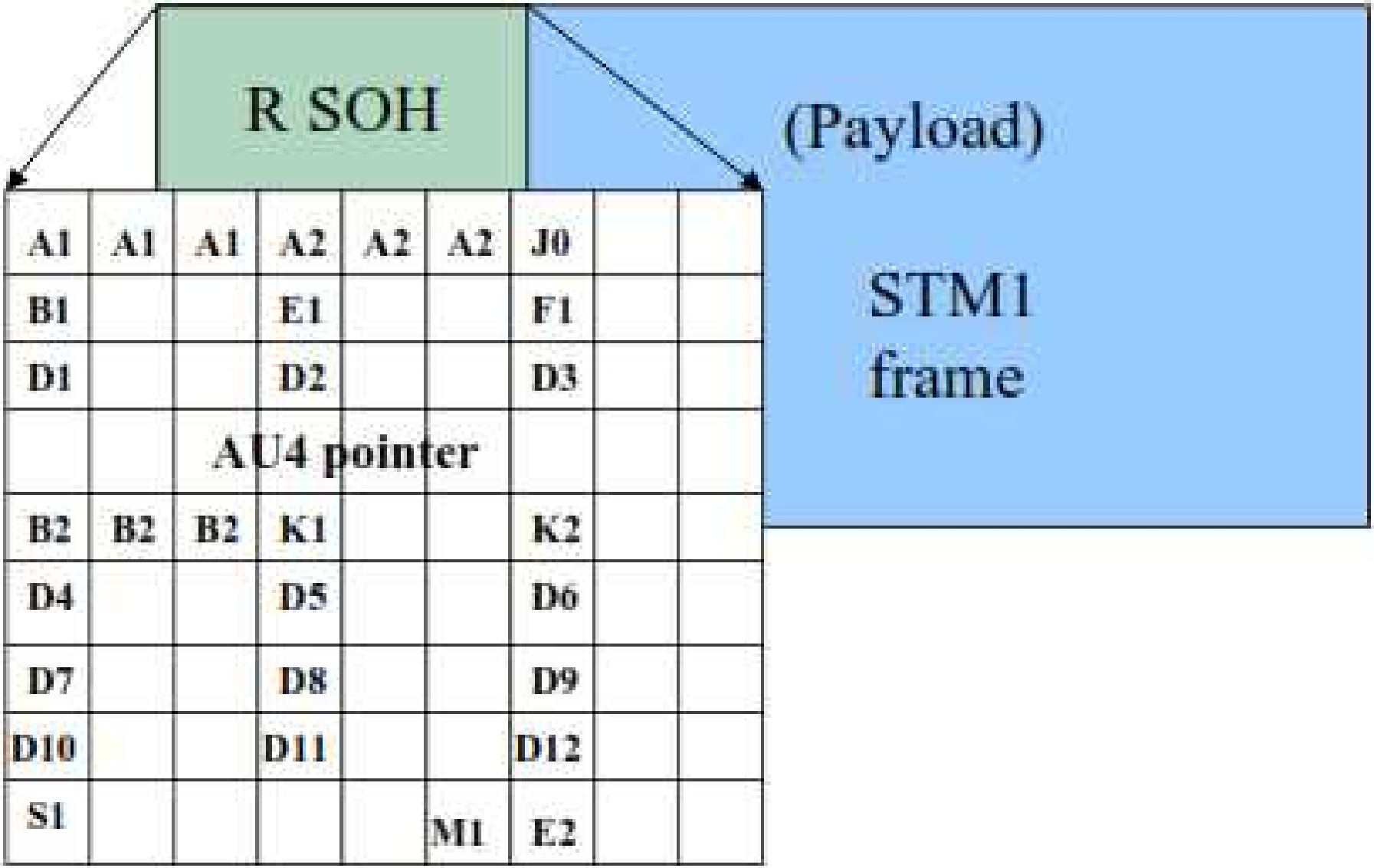
A1	A1	A1	A2	A2	A2	C1				J1
B1			E1			F1				B3
D1			D2			D3				C2
H1	H1	H1	H2	H2	H2	H3	H3	H3		G1
B2	B2	B2	K1			K2				F2
D4			D5			D6				H4
D7			D8			D9				Z3
D10			D11			D12				Z4
Z1	Z1	Z1	Z2	Z2	Z2	E2				Z5

A1, A2 – Framing
H1-H3 – AU (Administrative Unit) Pointers
B1 – BIP-8 (Bit interleaved parity)
B2 – BIP-24
C1, C2 – STM Identifiers
D1 to D12 - Data Communication Channels

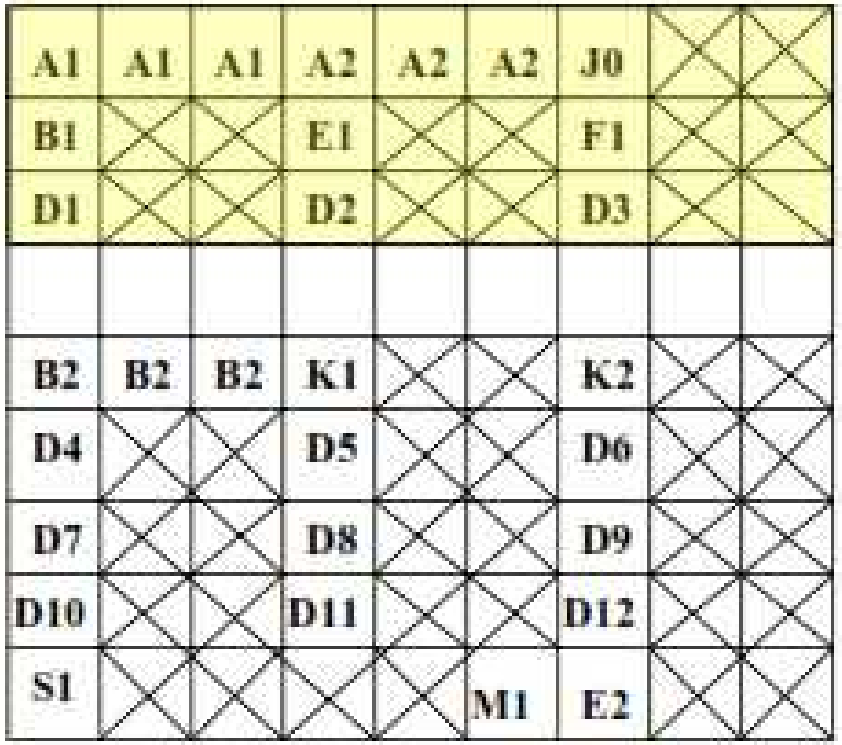
Section OH (SOH)
Line OH (LOH)
Path OH (POH)

E1, E2 - Order Wire
F1, F2 – User defined
G1 – Path status
J1 – Path trace
K1, K2 – Automatic protection Switching
Z1 to Z5 – Growth (reserved space)

Transport Overhead : SOH



Transport Overhead : SOH



A1 and A2 : frame alignment word

B1 : Regenerator setion error monitoring

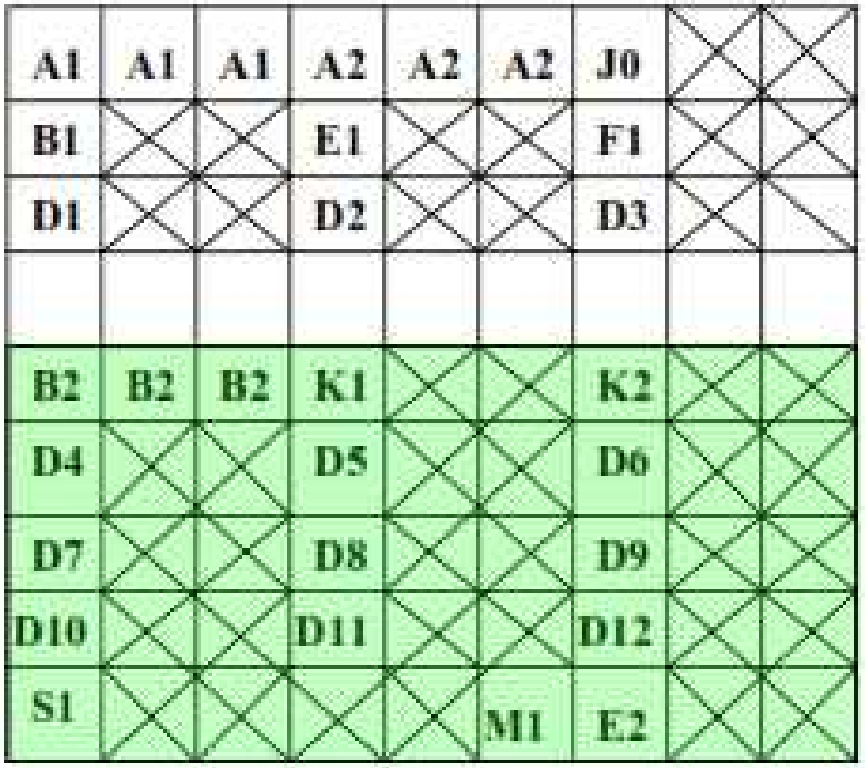
J0 : STM1 identifier (16 bytes word)

E1 : Service channel (transport a 64 Kbit/s channel)

F1 : user channel. May be used for network exploitation

D1-D3 : Data communication Channel at 192 Kbit/s.

Transport Overhead : SOH



B2 : Multiplexing section error monitoring

K1 and K2 : Automatic protection switching signalling

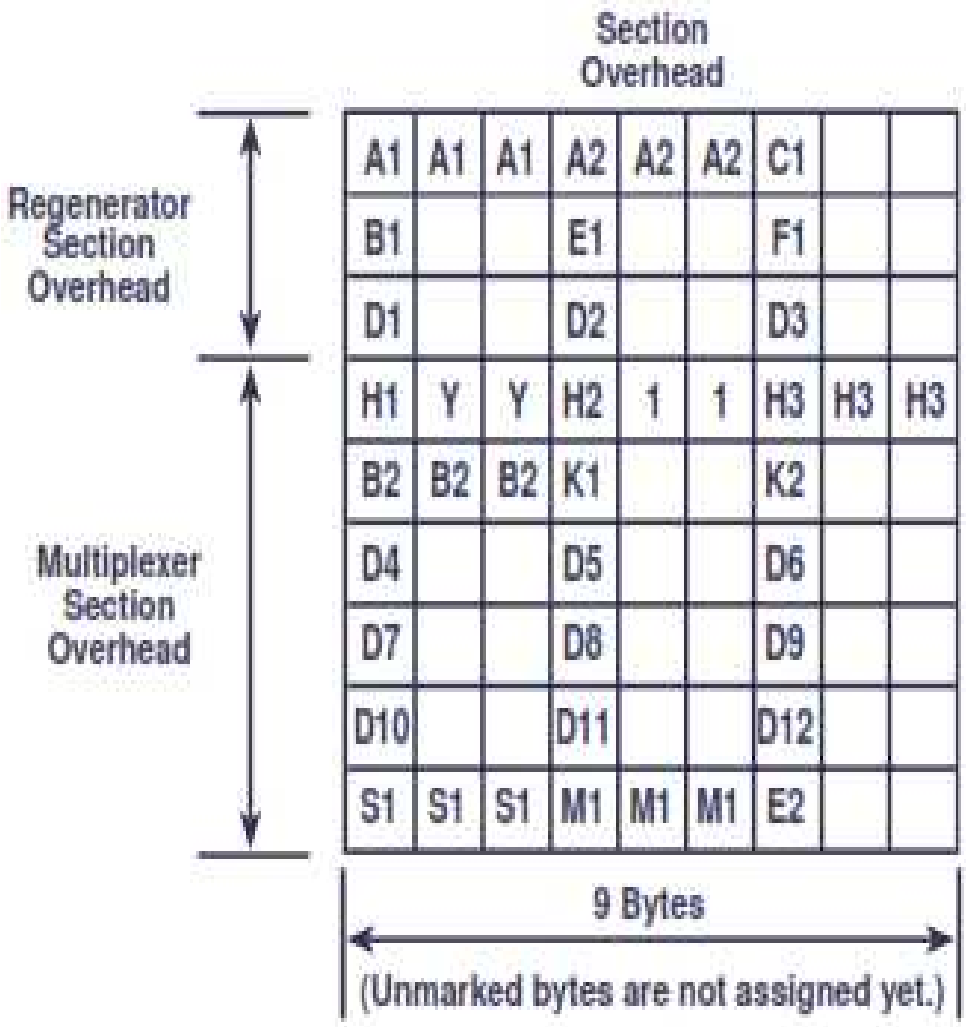
D4-D12 : Data communication Channel at 576 Kbit/s.

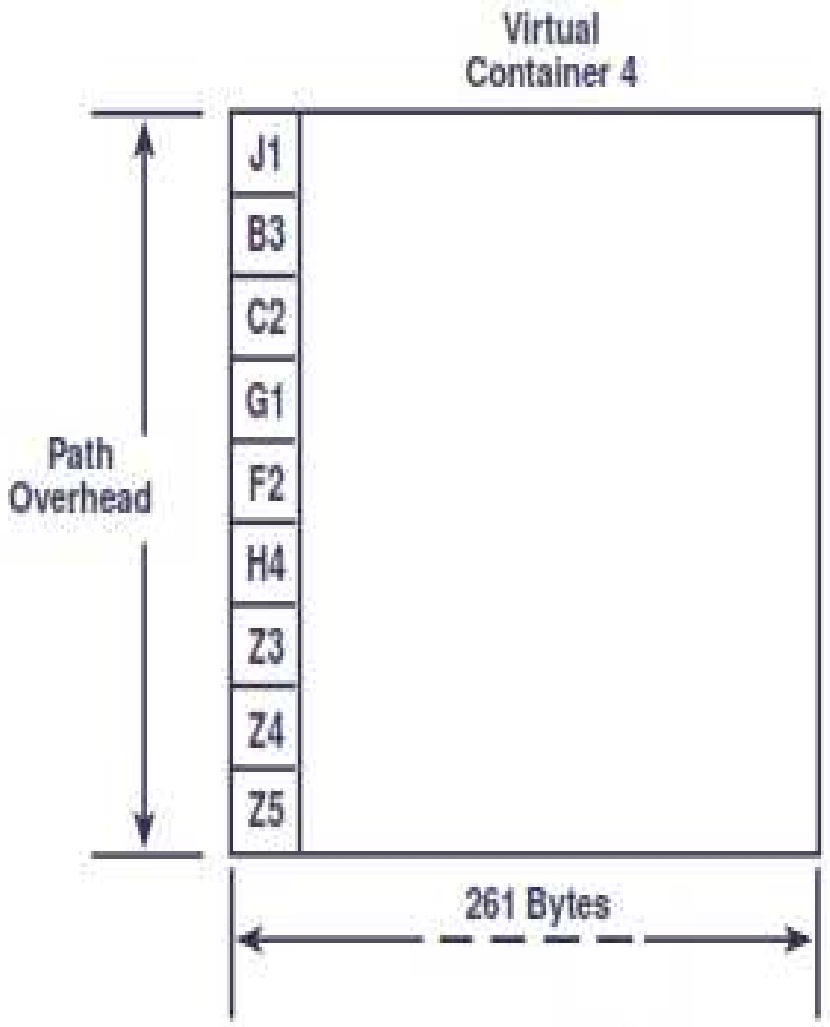
S1 : bytes of synchronization status

M1 : Binary code for number of errored blocks

E2 : service channel at 64 Kbit/s

SDH - Framing


Section Overhead
Regenerator Section Overhead
Multiplexer Section Overhead
A1 A1 A1 A2 A2 A2 C1
B1 E1 F1
D1 D2 D3
H1 Y Y H2 1 1 H3 H3 H3
B2 B2 B2 K1 K2
D4 D5 D6
D7 D8 D9
D10 D11 D12
S1 S1 S1 M1 M1 M1 E2
9 Bytes
(Unmarked bytes are not assigned yet.)


Virtual Container 4
Path Overhead
J1
B3
C2
G1
F2
H4
Z3
Z4
Z5
261 Bytes

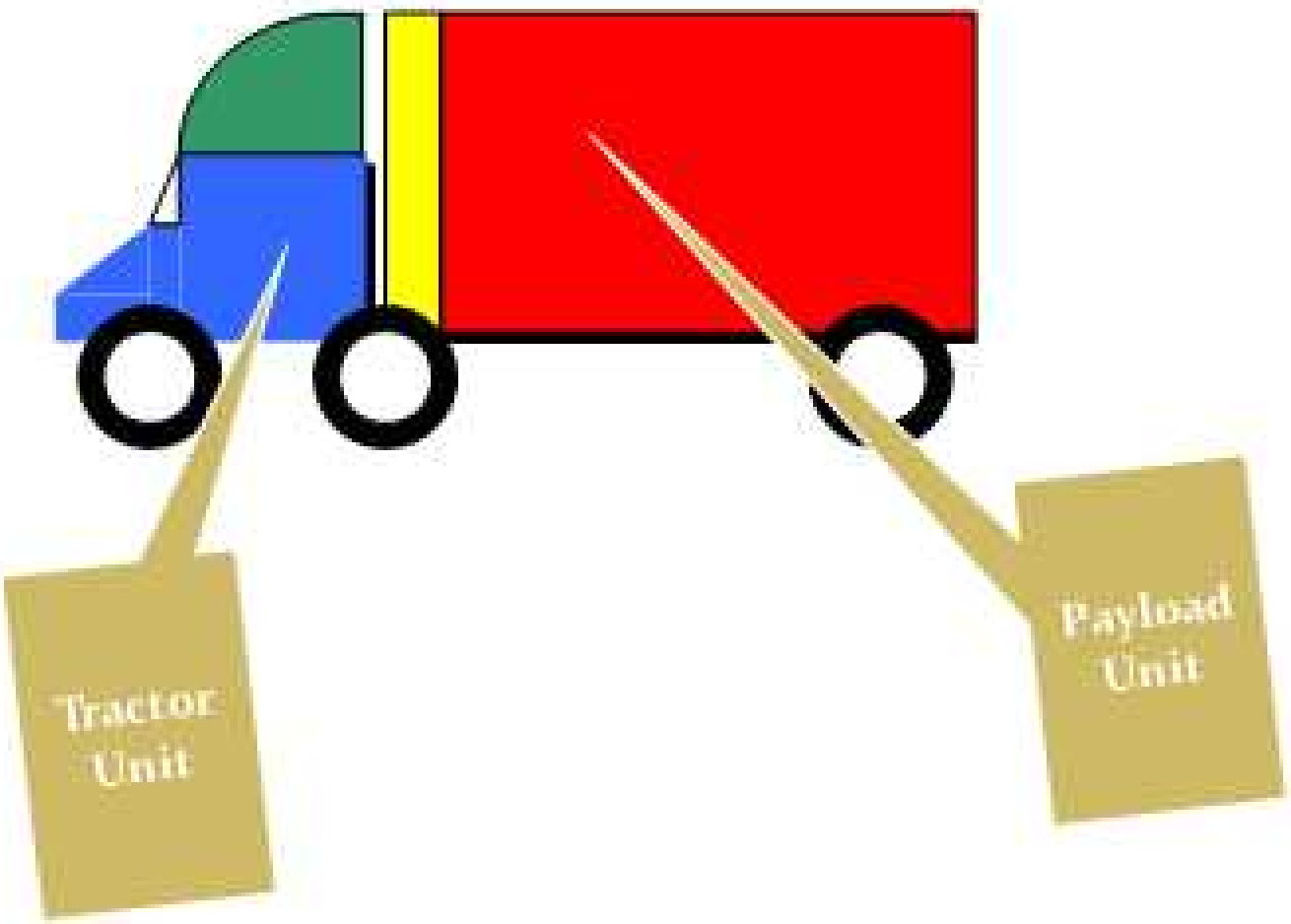
Transport Overhead : AU 4 pointer

A1	A1	A1	A2	A2	A2	J0		
B1			E1			F1		
D1			D2			D3		
B2	B2	B2	K1			K2		
D4			D5			D6		
D7			D8			D9		
D10			D11			D12		
S1					M1	E2		

The secret to making SDH work is the payload pointer. The tributaries coming into a multiplexer may have been created with a clock running at a different speed. They are not necessarily aligned with each other or with the clock in the multiplexer. To resolve this problem, remember that this is a SYNCHRONOUS network, the SDH multiplexer finds the beginning of a frame for each tributary.

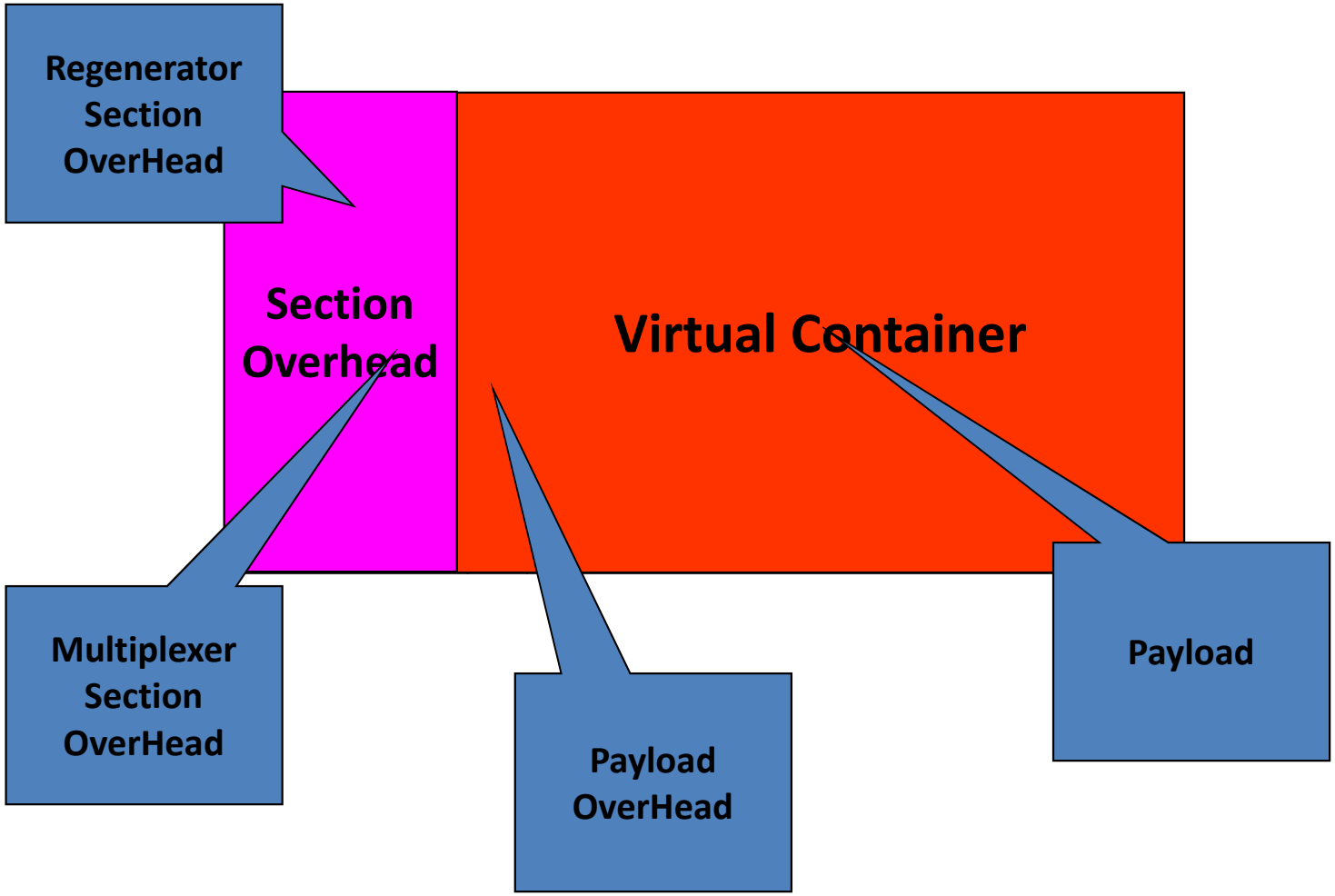
SDH FRAME STRUCTURE

TRUCK ANALOGY



SDH FRAME STRUCTURE

TRUCK ANALOGY contd.



Payload

- Isi container yang dibawa oleh truk mewakili nilai riil.
- Payload ini dianalogikan sebagai lalulintas costumer (data), yang sedang dibawa oleh container dalam sebuah SDH frame.
- Payload ini mensuport sesi transportasi dari spesifik tributary sinyal.

Section Overhead

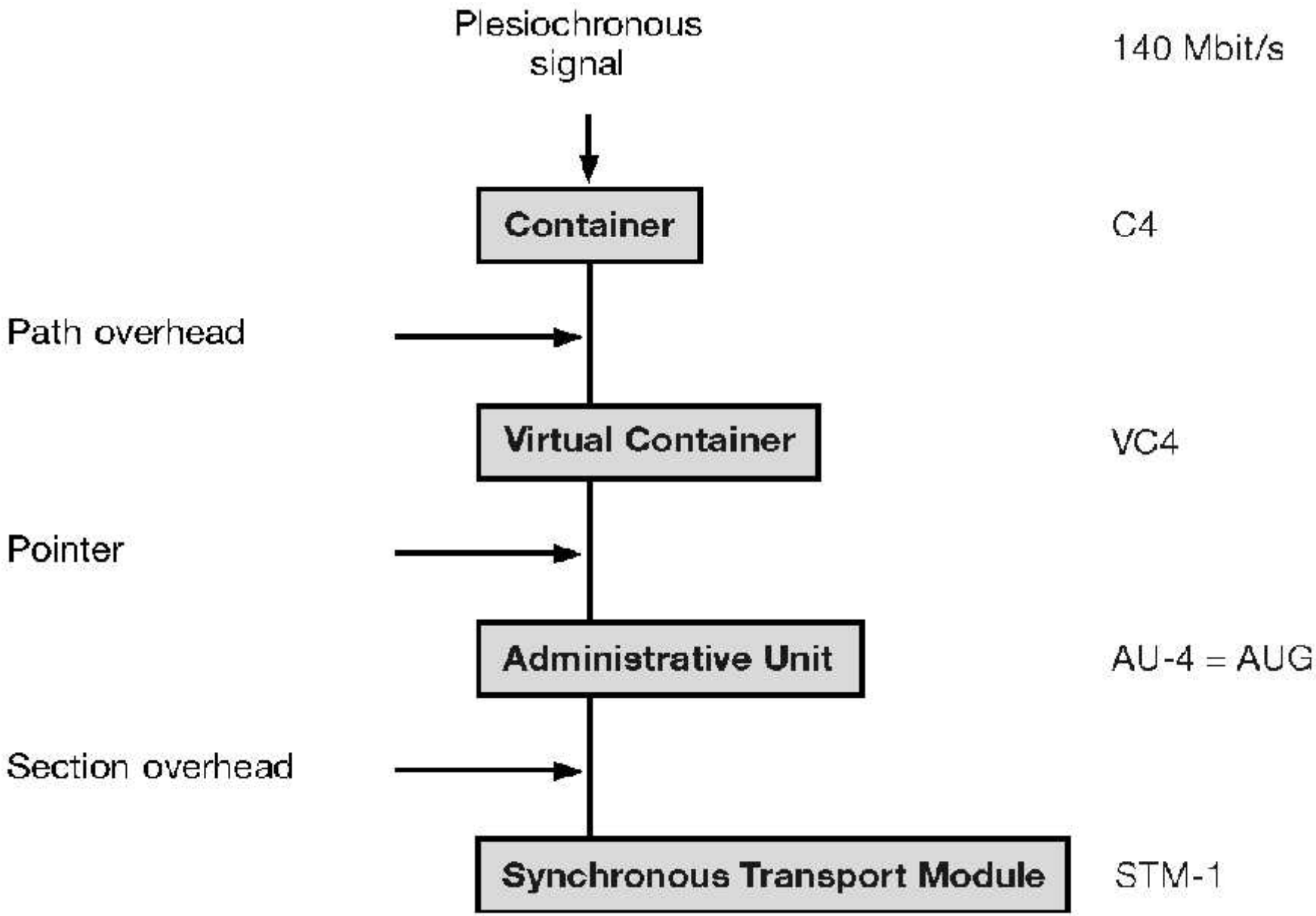
- Bagian yang mengetahui tujuan dari isi truk tersebut, yg pada gambar disebut tractor unit.
- Dianalogikan dengan pemeliharaan jaringan dan kemampuan manajemen yang dibawa oleh frame SDH, dikenal sebagai Section Overhead, atau SOH.

- Section Overhead (SOH) menyediakan fasilitas yang diperlukan untuk mendukung dan mempertahankan transportasi lalu lintas pelanggan yang aman di seluruh jaringan.
- SOH dibagi menjadi Multiplexer Section Overhead (MSOH) dan Regenerator Section Overhead (RSOH).

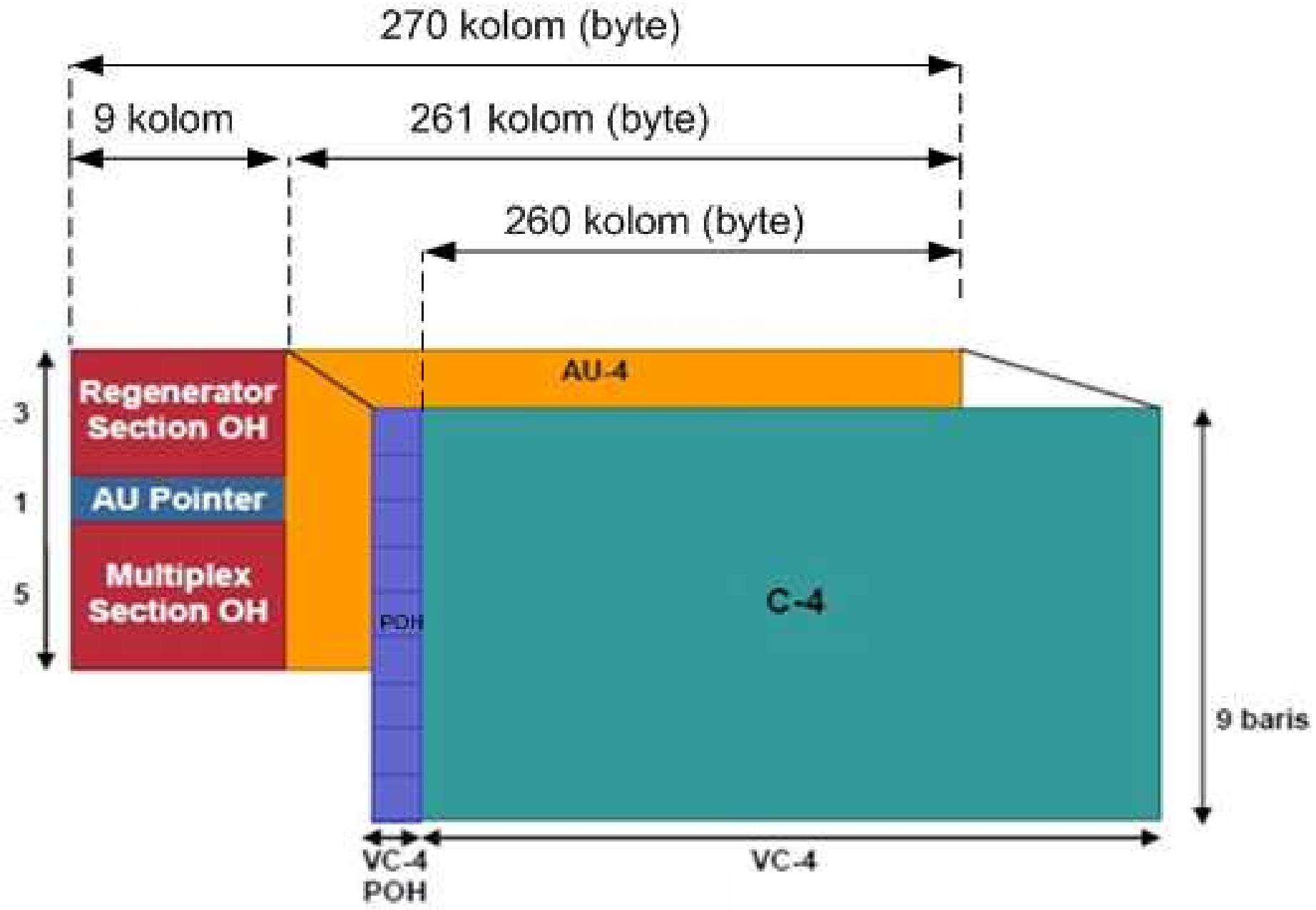
The Virtual Container

- Walaupun container dimuat oleh truk yang berbeda, ada sebagian overhead yang selalu tetap sama.
- Hal ini disebut juga Path OverHead, atau POH.
- Path Overhead berhubungan langsung dengan kapasitas area payload, mereka bergabung bersama membentuk apa yang kita ketahui sebagai Virtual Container.

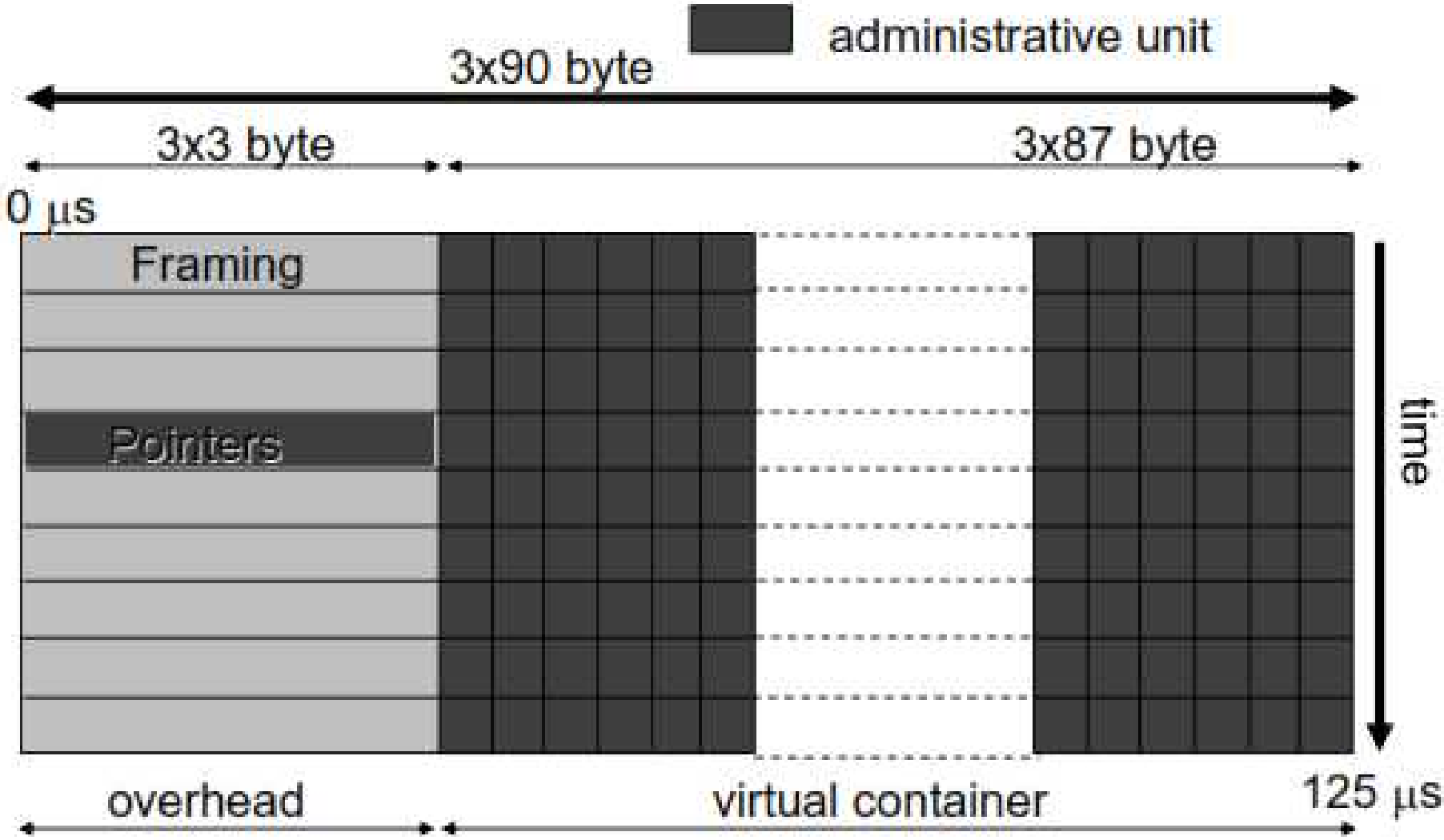
❖ Diaram Pembentukan STM-1



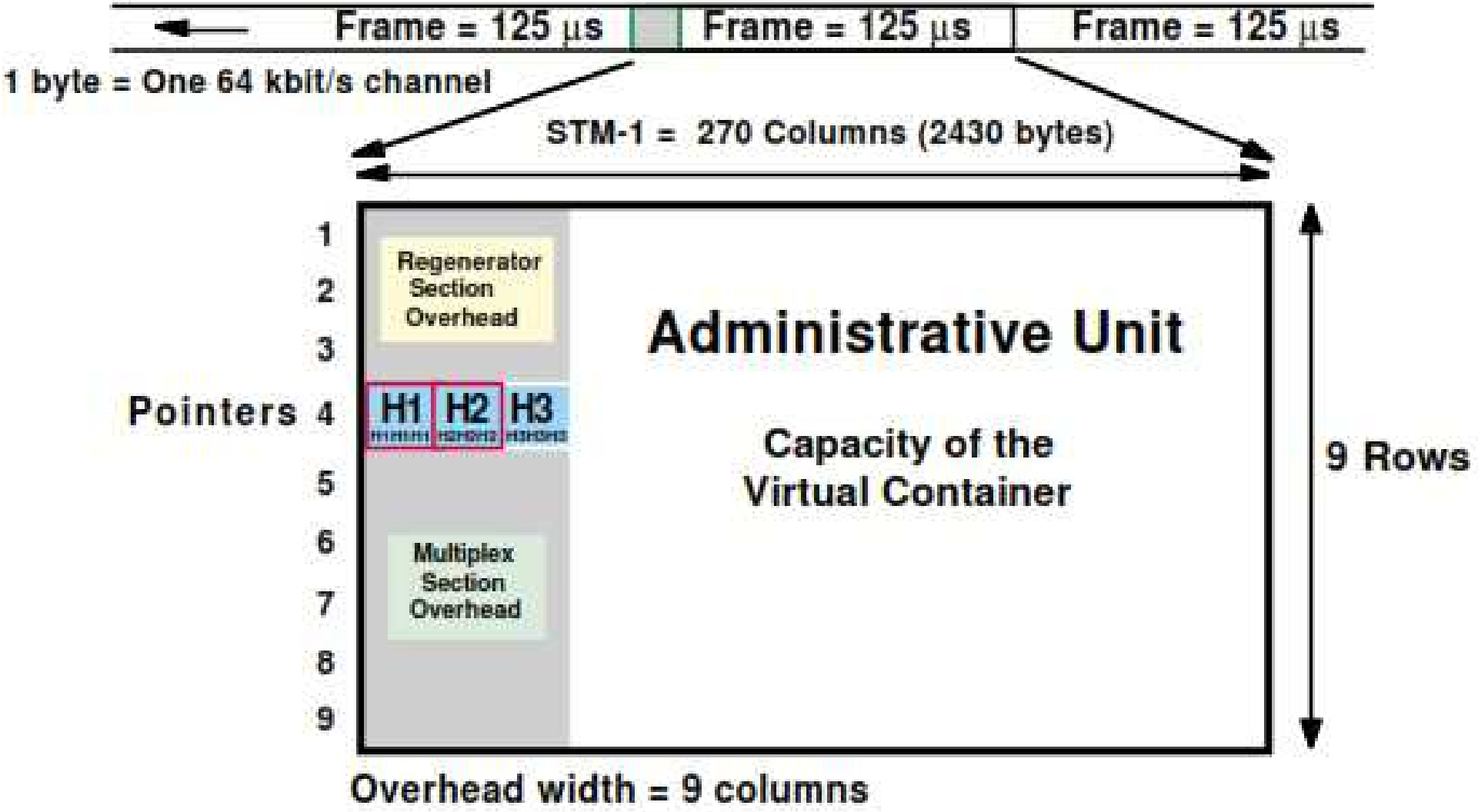
❖Struktur Frame STM-1



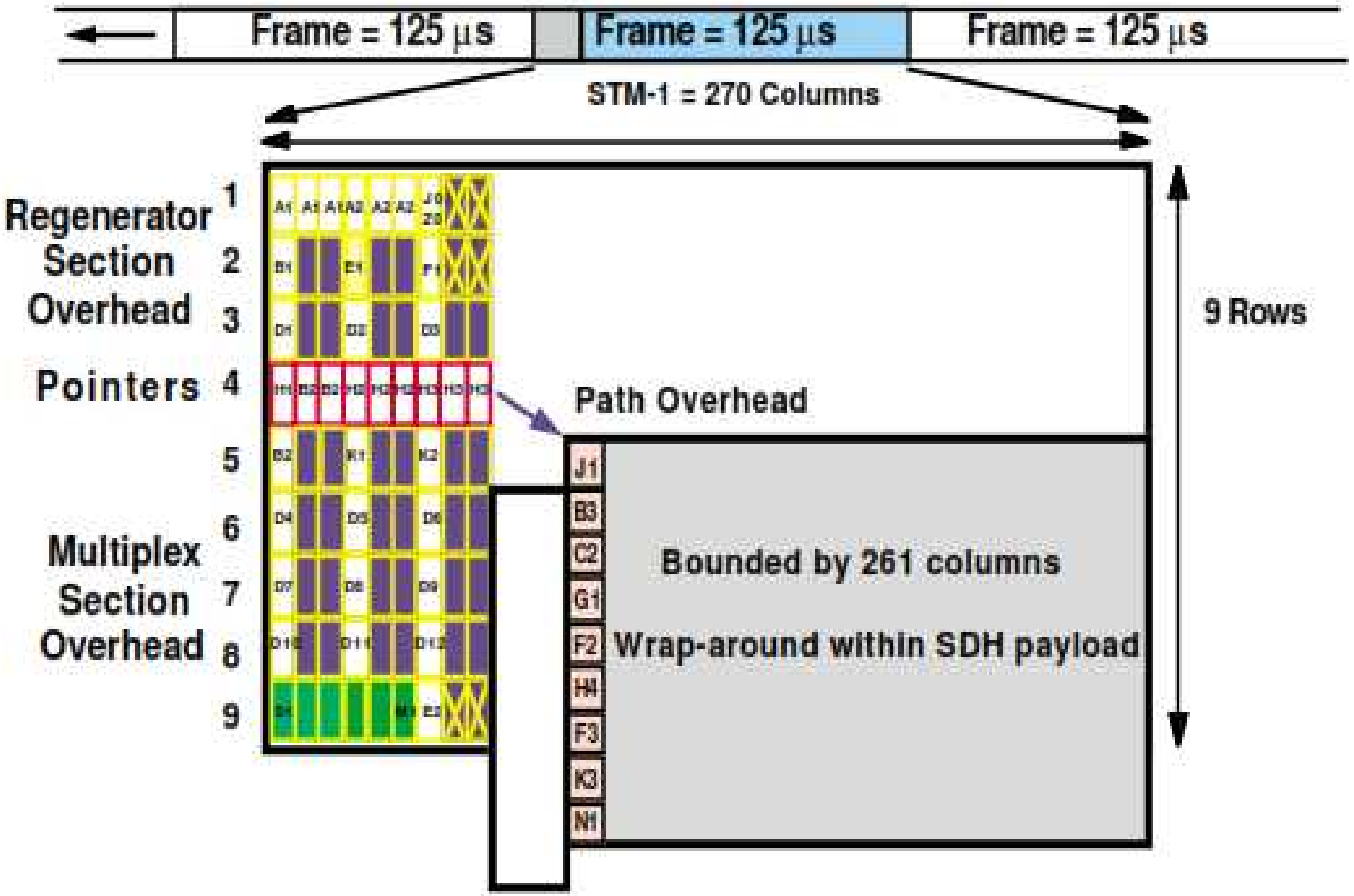
STM-1 frame in SDH



STM-1 frame in SDH



STM-1 frame in SDH



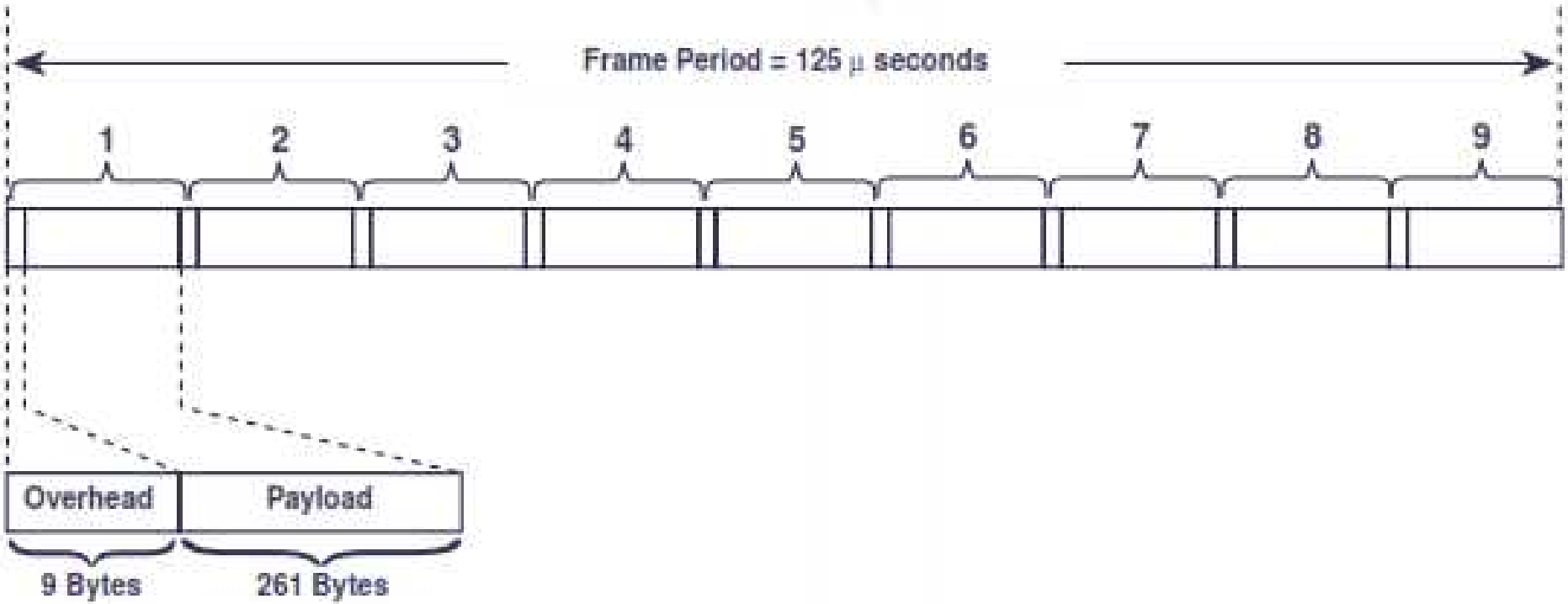
In order for SDH to easily integrate existing digital services into its hierarchy, it operates at the basic rate of 8 kHz, or 125 microseconds per frame, so the frame rate is 8,000 frames per second. The frame capacity of a signal is the number of bits contained within a single frame. ***Figure 1*** shows:

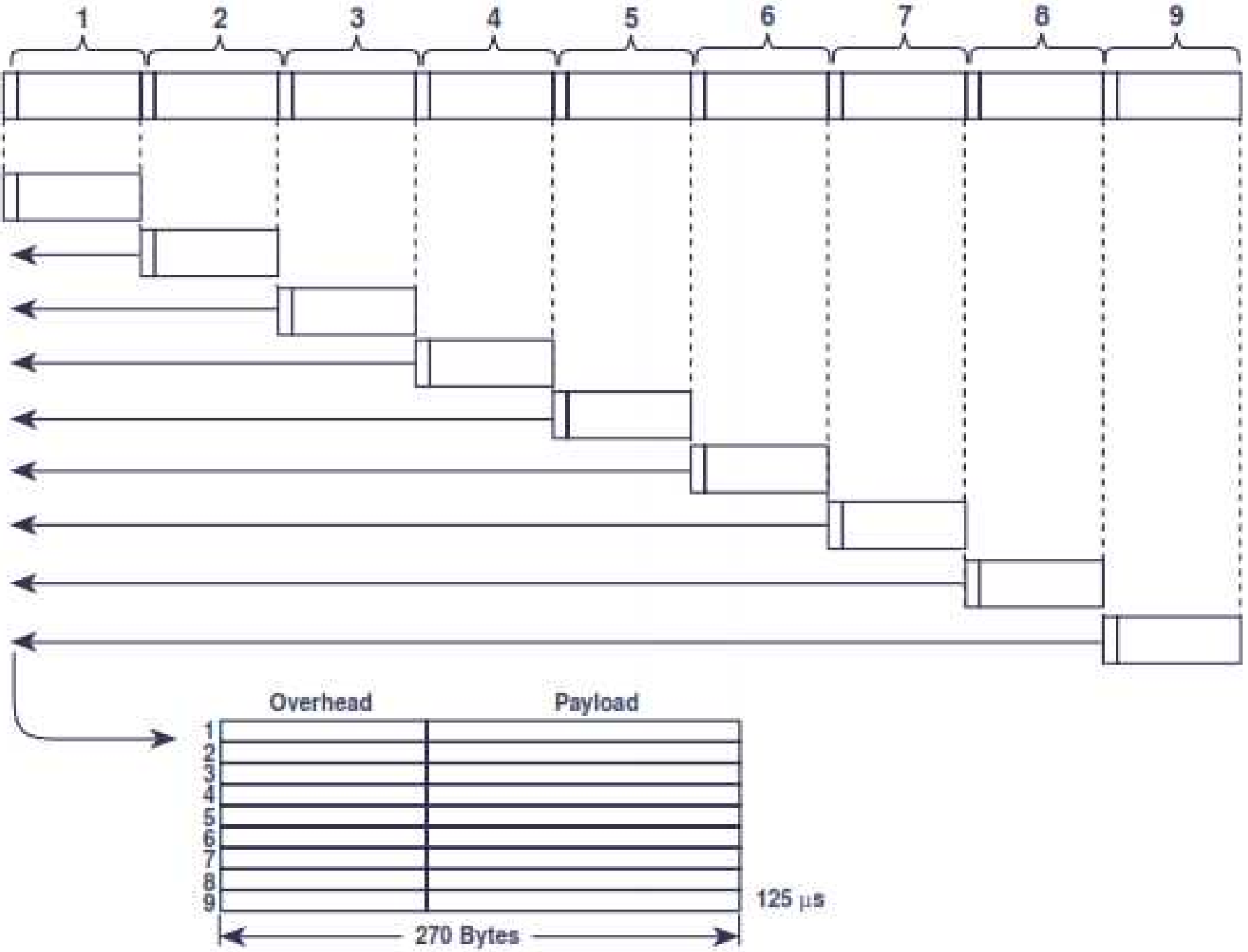
frame capacity = 270 bytes/row x 9 rows/frame x 8 bits/byte = 19,440 bits/frame

The bit rate of the STM-1 signal is calculated as follows:

bit rate = 8,000 frames/second x 19,440 bits/frame

= 155.52 Mbit/s

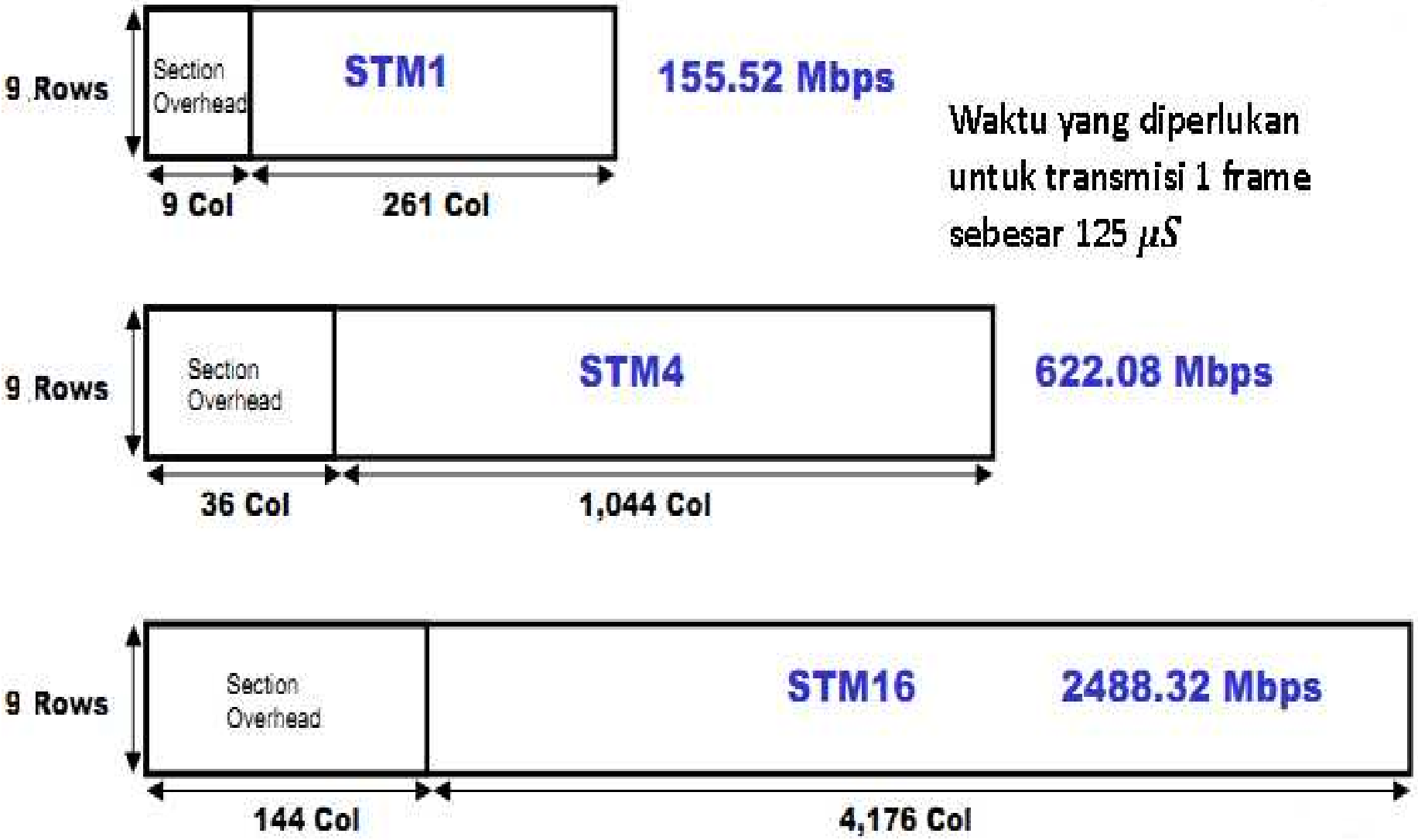

Frame Period = 125 μ seconds
1
2
3
4
5
6
7
8
9
Overhead
Payload
9 Bytes
261 Bytes


1
2
3
4
5
6
7
8
9
Overhead
Payload
1
2
3
4
5
6
7
8
9
125 µs
270 Bytes

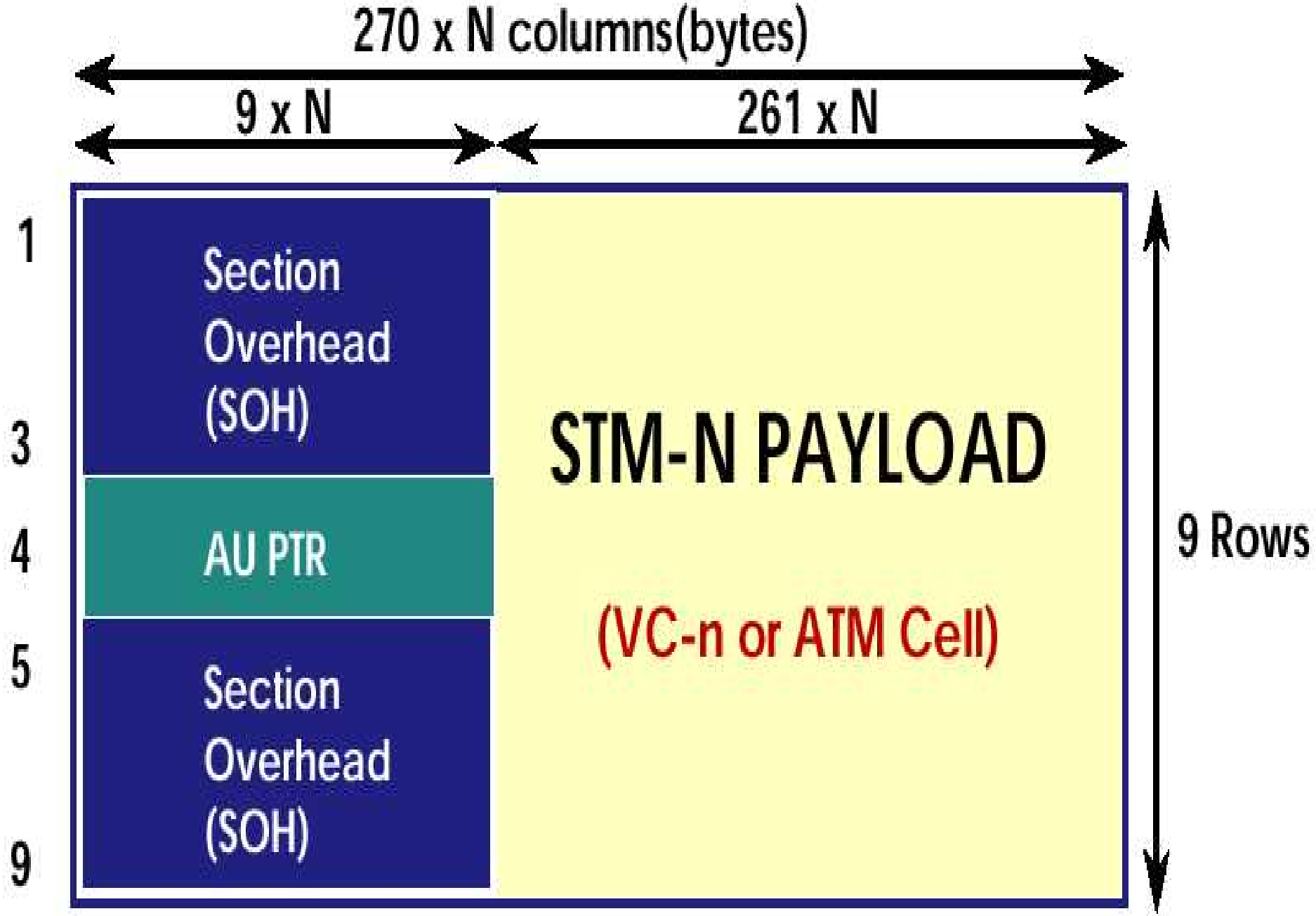
❖ SDH – Multiplexing



SDH – Multiplexing

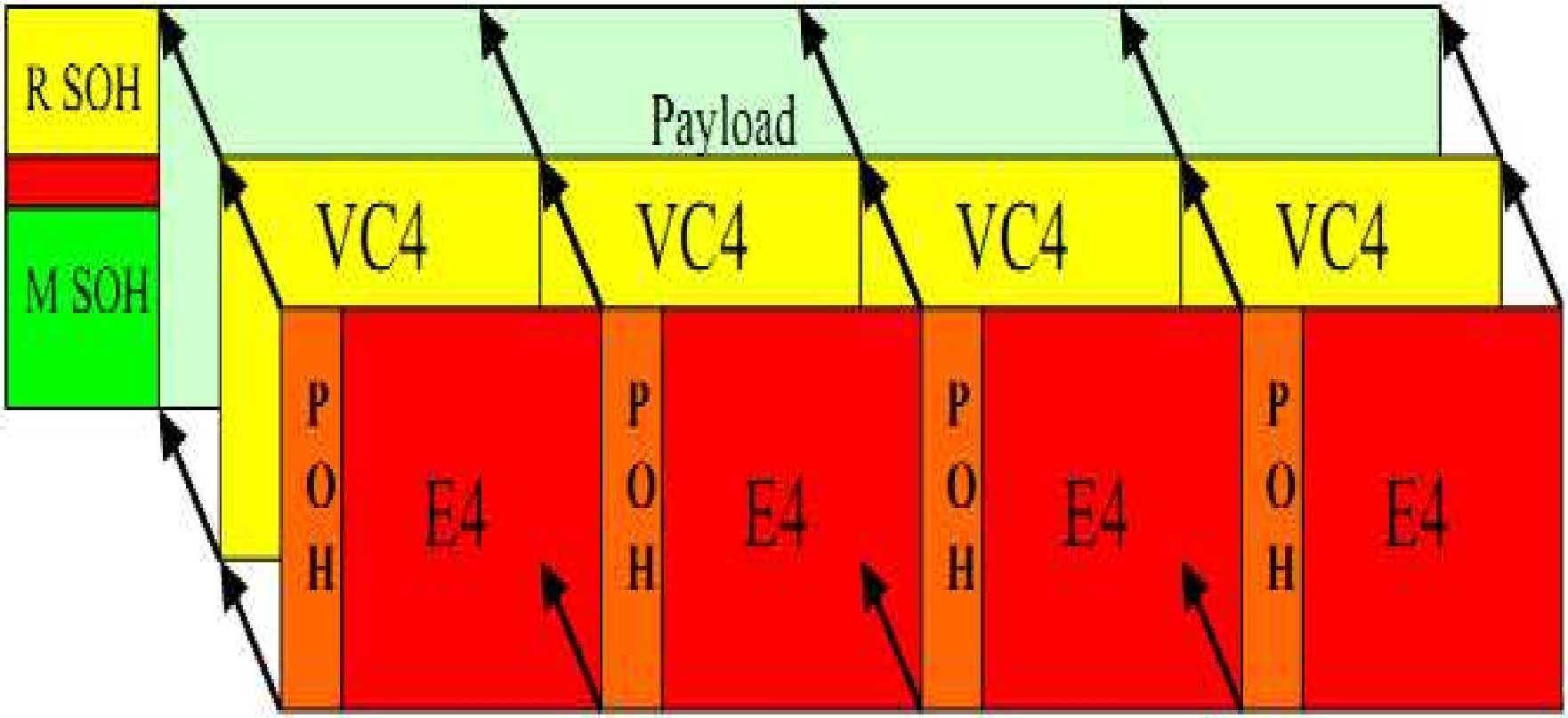


SDH – Multiplexing

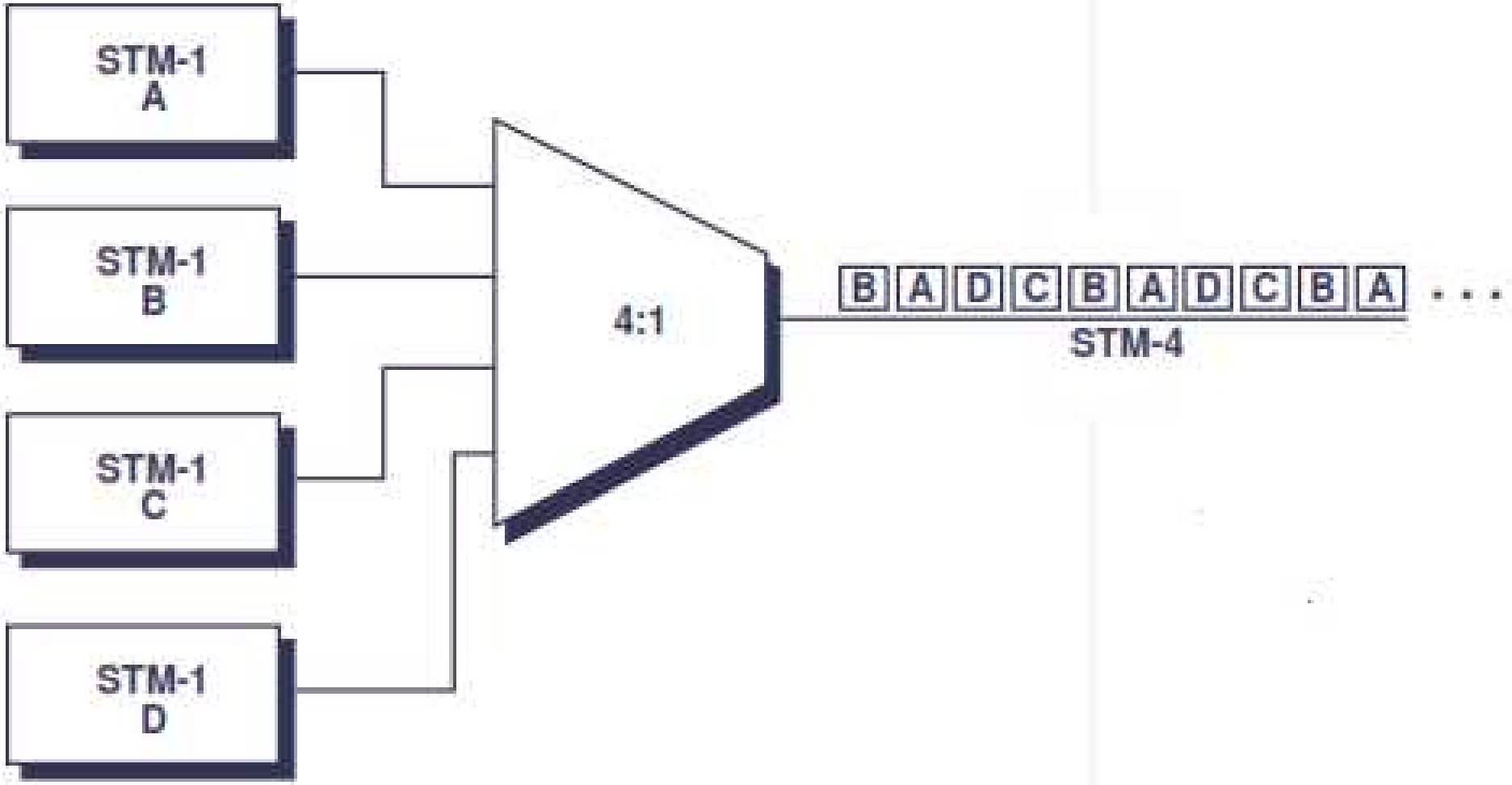


SDH – Multiplexing

- Susunan-susunan Frame STM-4

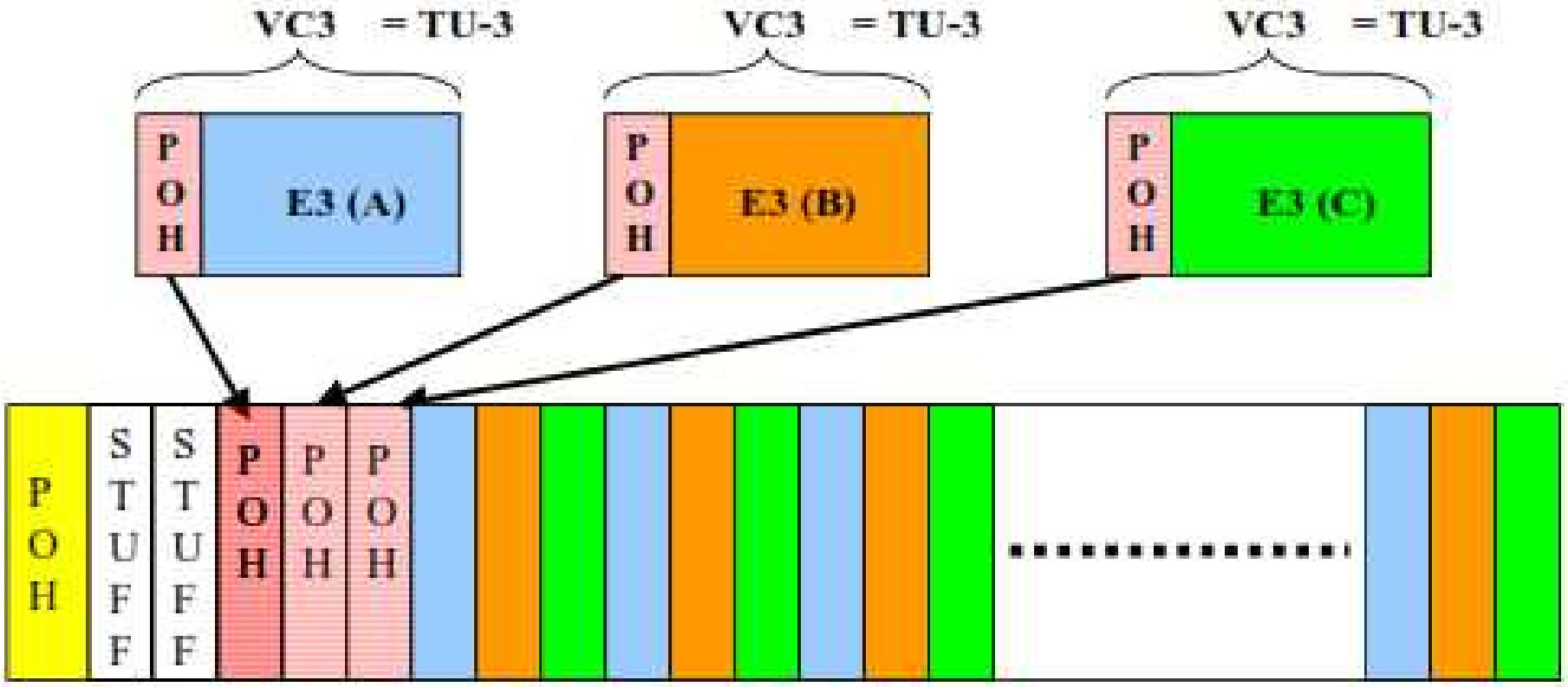


SDH – Multiplexing



SDH Multiplexing : another example (E3=>STM1)

You can map 3 E3 (34 Mbit/s) onto one STM1.



❖**The basic unit of SDH is STM-1:**

STM-1 = 155.52 Mbit/s

STM-4 = 622.08 Mbit/s

STM-16 = 2588.32 Mbit/s

STM-64 = 9953.28 Mbit/s

STM-256 = 39813.12 Mbit/s

STM-1024 = 159252.48 Mbit/s

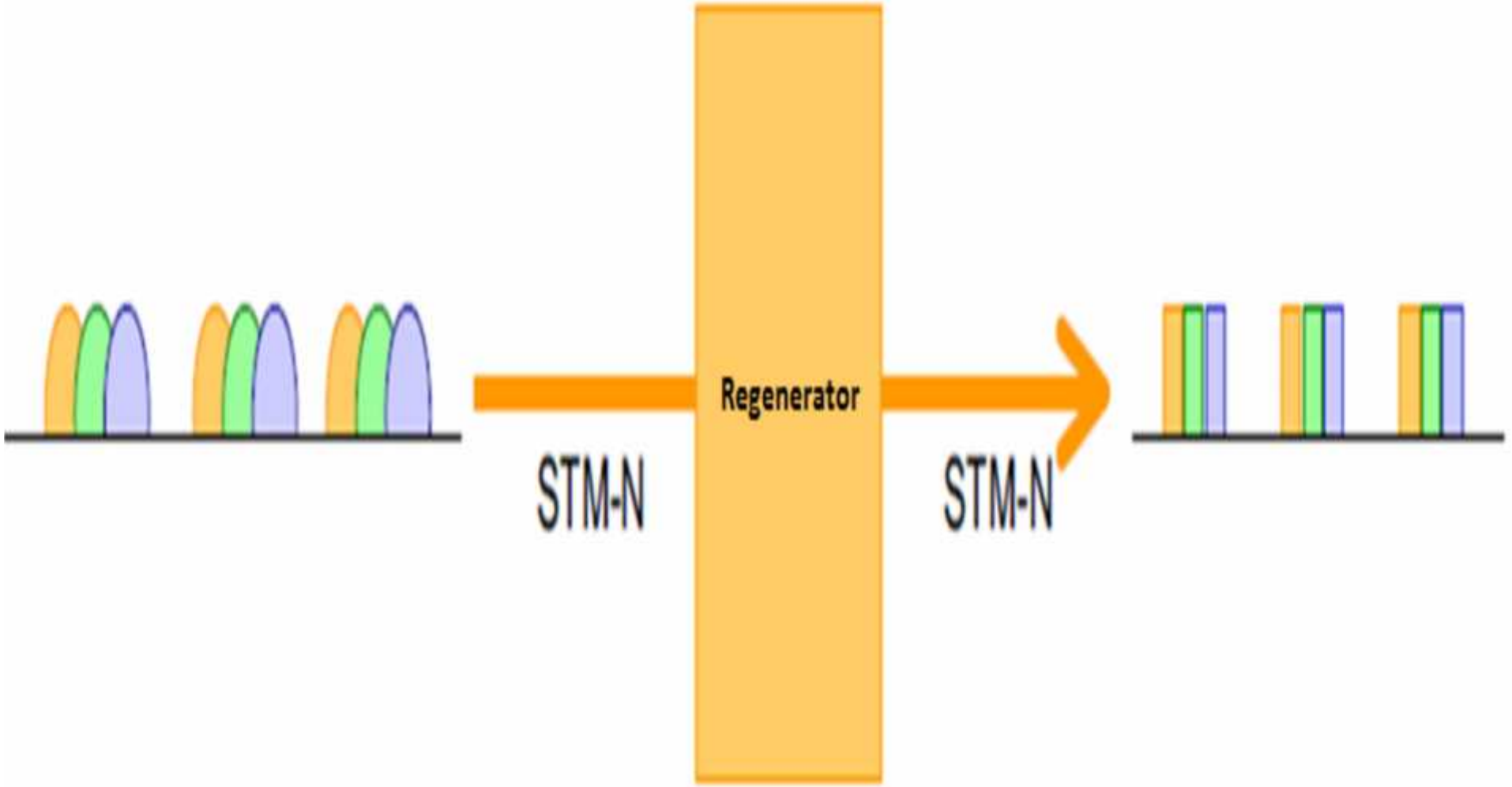
Elemen SDH

- Regenerator
- Terminal multiplexer (TM)
- Add(insert)/drop multiplexer (ADM)
- digital cross-connect (DXC)

Regenerator

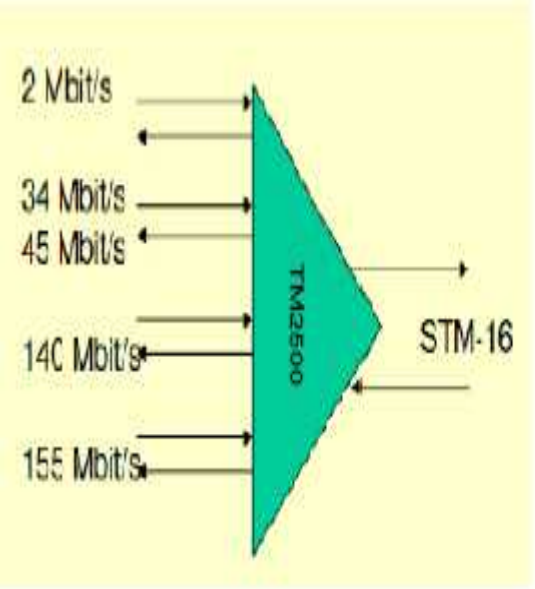
- Pada sistem transmisi digital, penguatan sinyal dilakukan menggunakan perangkat yang disebut *regenerator* (digital amplifier)
- Suatu *regenerator* terdiri dari *equalizing amplifier*, yang mengkompensasi distorsi dan menapis (mem-filter) *out-of-band noise*, serta sebuah komparator
- Sebuah regenator juga mengandung rangkaian pewaktu (timing) yang berfungsi untuk membangkitkan sinyal clock berdasarkan sinyal input yang diterima
- Sinyal hasil regenerasi akan bebas dari noise dan siap ditransmisikan lagi

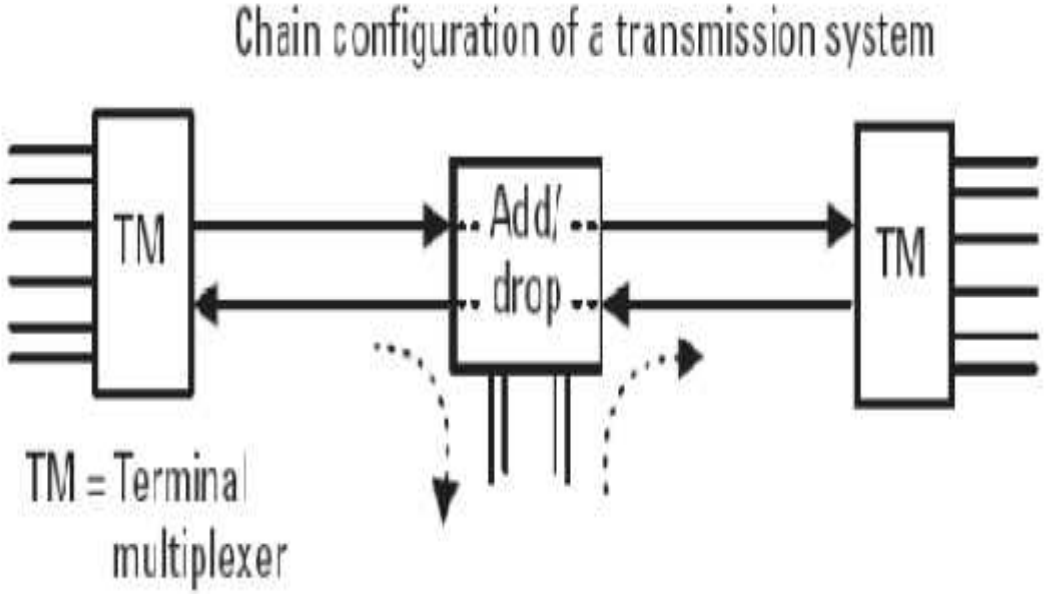
Regenerator



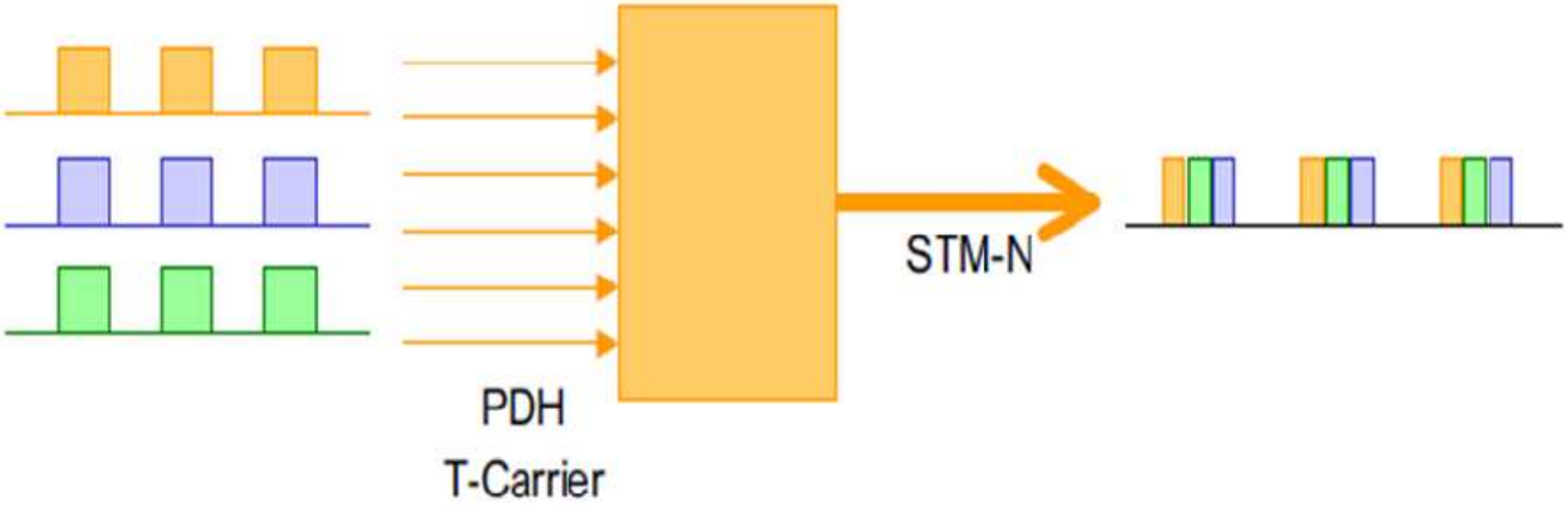
Terminal Multiplexer

- *Terminal multiplexer* (TM) atau *multiplexer* (saja) berfungsi untuk menggabungkan sinyal digital dengan tujuan memperoleh bit rate yang lebih tinggi untuk transmisi berkapasitas tinggi



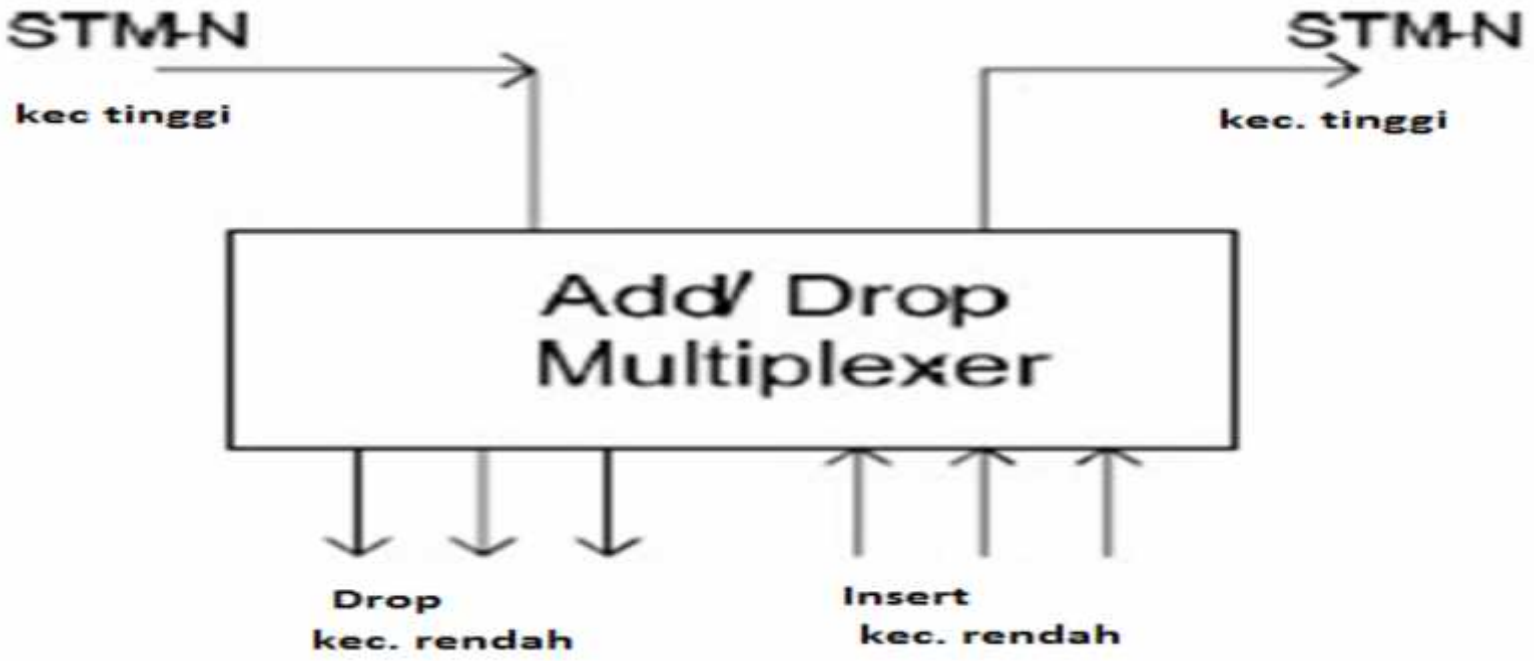


Terminal Multiplexer

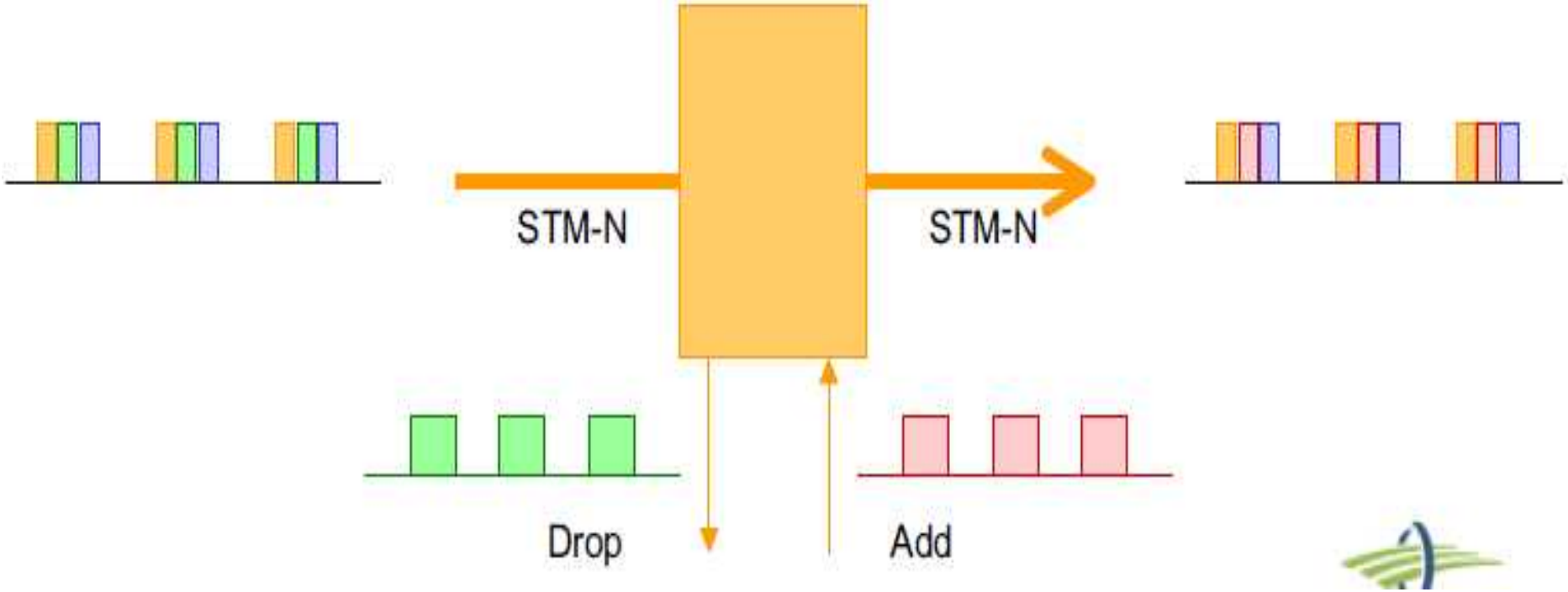


Add/Drop Multiplexer

- *Add/drop multiplexers* digunakan untuk mengambil (drop) beberapa kanal dari aliran data kecepatan tinggi atau untuk menyisipkan (add) beberapa kanal ke dalam aliran data berkecepatan tinggi



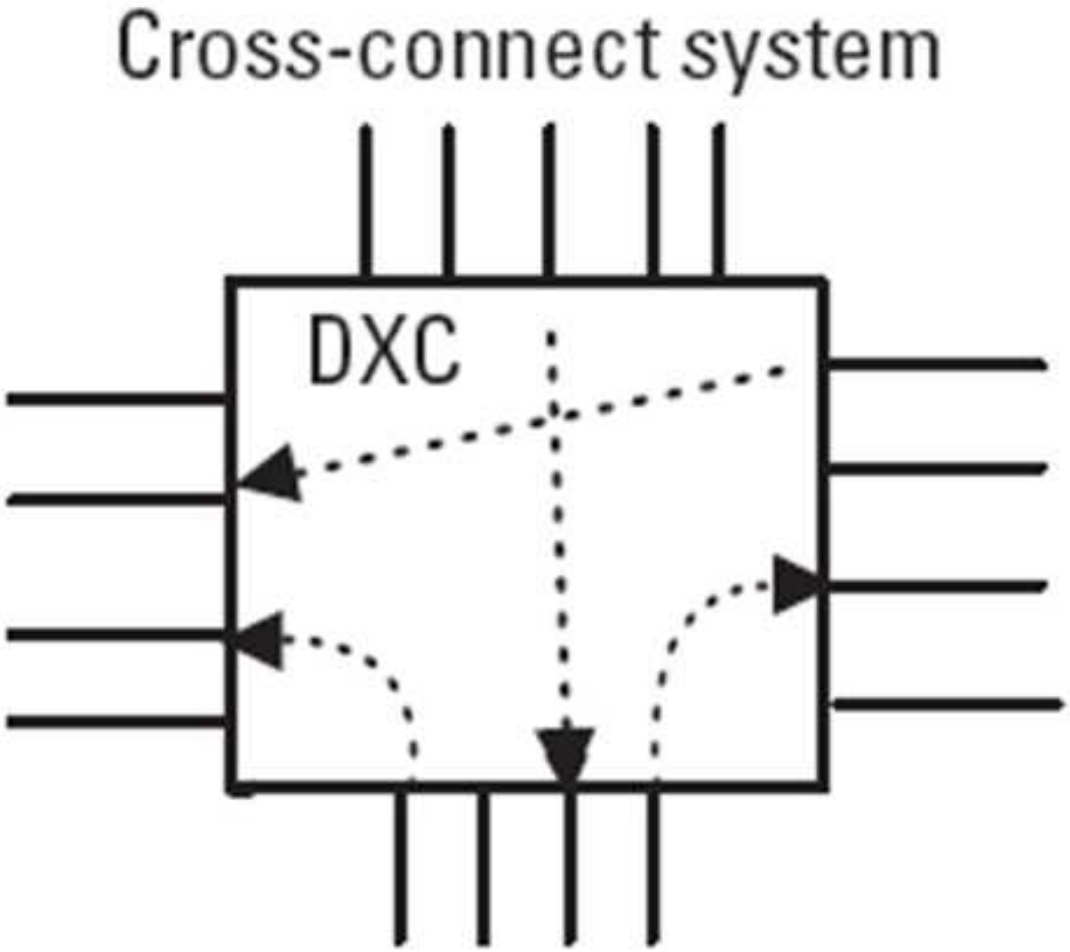
Add/Drop Multiplexer



Digital Cross-Connect

- *Digital cross-connect* (DXC) merupakan node jaringan yang mampu menyusun ulang kanal-kanal yang ada di dalam suatu aliran
- DXC memungkinkan konfigurasi terhadap jaringan dilakukan secara flexible
- Fungsi dasar DXC adalah sama dengan sentral
- DXC mampu men-switch pada orde tinggi (tidak hanya orde 64 Kbps seperti pada sentral biasa)
- DXC bisa jadi mengandung fungsi *redundancy* yang dapat secara otomatis mem-bypass bagian link transmisi yang rusak
- SDH dan SONET sering menggunakan topologi ring untuk mendapatkan keandalan (reliability) yang lebih tinggi

Digital Cross-Connect



Digital Cross-Connect

